

# claude-windows / 144a31b7-7962-4fee-9d6f-8a7854daa63f

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\144a31b7-7962-4fee-9d6f-8a7854daa63f\subagents\workflows\wf_887fb14e-e6d\agent-a0190d6f56c057865.jsonl`  
- SHA-256: `fbf113e54d19d828247bc6013b9ac6a90eecf7323021e1f871bf1fb838aeb597`  
- Source modified: `2026-06-20T10:05:45+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_887fb14e-e6d`  
- session_id: `144a31b7-7962-4fee-9d6f-8a7854daa63f`
```

## Transcript

### 1. user (2026-06-20T10:03:00.497Z)

Goal: design dedicated Studio-UI pages that turn "annotate your footage ? train your OWN model" into a first-class USP (the bring-your-own-model moat), linking the public GitHub repo (<https://github.com/avidullu/rally-annotator>), with screenshots and step-by-step setup/use help. Be HONEST about what is real today vs roadmap. Tag claims [verified]/[design]/[researched]. Do NOT invent capabilities. EXCLUDE any .claude/worktrees/ paths.

Map the khelsutra UI surface for placing dedicated pages (repo at C:\Users\avidu\Projects\khelsutra; engine docs at C:\Users\avidu\Projects\badminton-highlight-indexer). Read khelsutra site/public/index.html (marketing site structure + brand CSS), web/ (the just-scaffolded SPA console + README), docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md + docs/PER\_RALLY\_SELECTION.md (the console). Read C:\Users\avidu\Projects\badminton-highlight-indexer/docs/PROJECT\_DOC\_TEMPLATE.md for the tracked-doc structure. Recommend WHERE the annotate / train-your-own-model / setup pages belong: marketing pages on the public site (educate/convert) vs app screens in the SPA console (operate) vs both ? with reasons. List routes/components to add + assets to reuse. Quote path:line. Return the schema.

### 2. assistant (2026-06-20T10:03:03.153Z)

I'll start by understanding the khelsutra UI surface and the engine docs. Let me pull the repos fresh and read the key files in parallel.

### 3. user (2026-06-20T10:03:06.895Z)

Already up to date.

---STATUS---

**master...origin/master**

?? .claude/

### 4. user (2026-06-20T10:03:07.258Z)

```
./.claude/launch.json  
./.gitattributes  
./.github/workflows/parity.yml  
./.github/workflows/test.yml  
./.gitignore  
./docs/PER_RALLY_SELECTION.md  
./docs/PORTABILITY.md  
./docs/README.md  
./docs/REQUEST_ACCESS_FORM.md  
./NEXT_STEPS.md  
./README.md  
./site/make_og.py  
./site/package.json
```

```

./site/public/capture.html
./site/public/index.html
./site/public/og.png
./site/README.md
./site/schema.sql
./site/scripts/droplet-smoke.sh
./site/scripts/parity-check.sh
./site/server.js
./site/src/assets.test.js
./site/src/deploy-config.test.js
./site/src/index.js
./site/src/index.test.js
./site/src/notify.js
./site/src/server.test.js
./site/src/store.js
./site/wrangler.toml
./web/.gitignore
./web/dist/assets/index-B4_llFaB.js
./web/dist/assets/index-BX1rPprB.css
./web/dist/index.html
./web/index.html
./web/package-lock.json
./web/package.json
./web/README.md
./web/src/api/client.test.ts
./web/src/api/client.ts
./web/src/api/types.ts
./web/src/App.tsx
./web/src/index.css
./web/src/lib/log.ts
./web/src/main.tsx
./web/tsconfig.json
./web/vite.config.ts

```

## 5. assistant (2026-06-20T10:03:09.195Z)

Let me read the key UI surface files in parallel.

## 6. user (2026-06-20T10:03:10.392Z)

```

1      # web/ ? KhelSutra Studio (operator console SPA)
2
3      Vite + React + TS + Tailwind v4. The browser UI that drives the rally indexer
engine**:
4      upload ? index ? pick rallies ? reel ? download. See `docs/PER_RALLY_SELECTION.md` (the
5      per-rally curation screen) and `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the single-box
appliance
6      this SPA fronts).
7
8      ## Dev
9      ```
10     npm install
11     npm run dev          # http://localhost:5173 ; proxies /api ? http://localhost:8000 (the
engine)
12     ```
13     Run the engine alongside it: `python -m backend.main --server` (in the badminton-
highlight-indexer
14     repo). The SPA uses same-origin relative `/api/*` calls ? in the appliance a reverse
proxy serves
15     this bundle and proxies `/api` to the engine on `127.0.0.1:8000`.
16
17     ## Checks (CI gates merge)
18     ```
19     npm run typecheck    # tsc --noEmit
20     npm run test         # vitest run (unit tests; e.g. the typed API client)

```

```

21     npm run build      # vite build
22     ``
23
24     ## Layout
25     - `src/api/client.ts` ? typed engine client (`listRallies` / `compileReel` /
`highlightUrl` / `getConfig`),
26     grounded in the verified engine contract; every call carries a correlation id.
27     - `src/api/types.ts` ? engine data types (honest: `server`/`receiver`/`events` are
motion-only
28     fabrications and must not be surfaced).
29     - `src/lib/log.ts` ? structured client logging + correlation ids (observability seed,
appliance $5).
30     - `src/App.tsx` ? the console shell (screens land in M2+).
31

```

## 7. user (2026-06-20T10:03:10.565Z)

```

1     <!doctype html>
2     <html lang="en">
3     <head>
4     <meta charset="utf-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1" />
6     <title>KhelSutra ? every rally indexed, the dead time gone</title>
7     <meta name="description" content="KhelSutra turns a full badminton match into a clean
index of rallies with the dead time stripped out ? a highlight reel you can watch and download.
Invite-only alpha, Bengaluru." />
8     <meta property="og:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
9     <meta property="og:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha for academies and players in Bengaluru." />
10    <meta property="og:type" content="website" />
11    <meta property="og:url" content="https://kheIsutra.guru" />
12    <meta property="og:image" content="https://kheIsutra.guru/og.png" />
13    <meta property="og:image:width" content="1200" />
14    <meta property="og:image:height" content="630" />
15    <meta name="twitter:card" content="summary_large_image" />
16    <meta name="twitter:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
17    <meta name="twitter:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha, Bengaluru." />
18    <meta name="twitter:image" content="https://kheIsutra.guru/og.png" />
19    <meta name="theme-color" content="#0C261F" />
20    <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'/%3E%3Cpath d='M13
10 L23 16 L13 22 Z' fill='%2319B36B'/%3E%3C/svg%3E" />
21    <link rel="preconnect" href="https://fonts.googleapis.com" />
22    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
23    <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;
700;800&display=swap" rel="stylesheet" />
24    <style>
25        :root{
26            --ink:#0C261F; --ink2:#123A2E; --green:#19B36B; --green-d:#149A5C;
27            --lime:#C7F2B0; --paper:#F6F8F4; --card:#fff;
28            --tx:#14271F; --mut:#566B61; --line:#E4E9E2;
29            --on-dark:#EAF3EE; --on-dark-mut:#A7C3B7;
30            --r:14px; --rs:10px; --maxw:1080px;
31            --font:'Plus Jakarta Sans',system-ui,-apple-system,Segoe
UI,Roboto,Helvetica,Arial,sans-serif;
32        }
33        *{box-sizing:border-box}
34        html{scroll-behavior:smooth}
35        body{margin:0;font-family:var(--font);color:var(--tx);background:var(--paper);line-
height:1.65;-webkit-font-smoothing:antialiased}
36        a{color:inherit}
37        .wrap{max-width:var(--maxw);margin:0 auto;padding:0 22px}

```

```

38     h1,h2,h3{line-height:1.15;margin:0;letter-spacing:-.02em}
39     h1{font-size:clamp(2rem,5vw,3.4rem);font-weight:800}
40     h2{font-size:clamp(1.5rem,3.2vw,2.2rem);font-weight:700}
41     h3{font-size:1.15rem;font-weight:700}
42     p{margin:.6rem 0}
43     .btn{display:inline-flex;align-items:center;gap:.5rem;font-weight:600;font-
size:1rem;border-radius:999px;padding:.78rem 1.4rem;text-decoration:none;border:1px solid
transparent;cursor:pointer;transition:.15s}
44     .btn-primary{background:var(--green);color:#04140D}
45     .btn-primary:hover{background:var(--green-d)}
46     .btn-ghost{background:transparent;border-color:rgba(255,255,255,.28);color:var(--on-
dark)}
47     .btn-ghost:hover{border-color:rgba(255,255,255,.6)}
48     .btn-dark{background:var(--ink);color:#fff}
49     .btn-dark:hover{background:var(--ink2)}
50     .eyebrow{display:inline-flex;align-items:center;gap:.5rem;font-size:.82rem;font-
weight:600;letter-spacing:.02em;
51         background:rgba(25,179,107,.14);color:#0a6c45;border:1px solid
rgba(25,179,107,.3);padding:.3rem .8rem;border-radius:999px}
52
53     /* header */
54     header{position:sticky;top:0;z-index:20;background:rgba(246,248,244,.86);backdrop-
filter:blur(10px);border-bottom:1px solid var(--line)}
55     .nav{display:flex;align-items:center;justify-content:space-between;height:66px}
56     .brand{display:flex;align-items:center;gap:.6rem;font-weight:800;font-
size:1.18rem;letter-spacing:-.02em;text-decoration:none}
57     .brand .dot{color:var(--green)}
58     .nav-links{display:flex;align-items:center;gap:1.6rem}
59     .nav-links a{font-size:.95rem;font-weight:500;color:var(--mut);text-decoration:none}
60     .nav-links a:hover{color:var(--tx)}
61     .brand .play{display:inline-block;width:0;height:0;border-style:solid;border-width:7px
0 7px 12px;border-color:transparent transparent transparent var(--green);margin:0
.32em;transform:translateY(1px)}
62
63     /* hero */
64     .hero{background:linear-gradient(160deg,var(--ink) 0%,var(--ink2)
100%);color:var(--on-dark);position:relative;overflow:hidden}
65     .hero .wrap{padding-top:74px;padding-bottom:88px;position:relative;z-index:2}
66     .hero .eyebrow{background:rgba(199,242,176,.12);color:var(--lime);border-
color:rgba(199,242,176,.25)}
67     .hero h1{margin:1.1rem 0 0;max-width:15ch}
68     .hero h1 .hl{color:var(--lime)}
69     .hero .lead{font-size:clamp(1.05rem,1.6vw,1.28rem);color:var(--on-dark-mut);max-
width:54ch;margin:1.2rem 0 0}
70     .hero-cta{display:flex;flex-wrap:wrap;gap:.8rem;margin-top:2rem}
71     .hero-note{margin-top:1.1rem;font-size:.86rem;color:var(--on-dark-mut)}
72     .shuttle{position:absolute;right:-60px;top:-40px;width:520px;max-
width:55%;opacity:.16;z-index:1;pointer-events:none}
73     .loop{display:flex;flex-wrap:wrap;gap:.6rem;margin-top:2.4rem}
74     .loop .chip{display:flex;align-
items:center;gap:.5rem;background:rgba(255,255,255,.06);border:1px solid rgba(255,255,255,.12);
75         border-radius:999px;padding:.5rem 1rem;font-weight:500;font-
size:.92rem;color:var(--on-dark)}
76     .loop .chip b{color:var(--lime);font-weight:700}
77     .loop .arrow{color:var(--on-dark-mut);align-self:center}
78
79     .hero-grid{display:grid;grid-template-columns:1.04fr .96fr;gap:46px;align-
items:center}
80     .hero-left{min-width:0}
81     .hero-visual{min-width:0}
82     .reel-card{background:#fff;border-radius:18px;border:1px solid
rgba(255,255,255,.14);box-shadow:0 26px 60px -24px rgba(0,0,0,.55);overflow:hidden}
83     .reel-head{display:flex;align-items:center;gap:.55rem;padding:.7rem .9rem;border-
bottom:1px solid var(--line)}
84     .reel-head .dots{display:flex;gap:5px}

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85     .reel-head .dots i{width:9px;height:9px;border-
radius:50%;background:#D9DED7;display:block}
86     .reel-head .file{font-size:.78rem;color:var(--mut);font-weight:600}
87     .reel-video{position:relative;aspect-ratio:16/9;background:linear-
gradient(150deg,#0E3A2C,#17533C);display:flex;align-items:center;justify-content:center}
88     .reel-video .court{position:absolute;inset:16px;border:2px solid
rgba(199,242,176,.30);border-radius:6px}
89     .reel-video .court:before{content:"";position:absolute;left:50%;top:0;bottom:0;width:2
px;background:rgba(199,242,176,.30)}
90     .reel-video .court:after{content:"";position:absolute;top:50%;left:0;right:0;height:2p
x;background:rgba(199,242,176,.22)}
91     .reel-video .play{width:60px;height:60px;border-
radius:50%;background:var(--lime);display:flex;align-items:center;justify-
content:center;color:#0C261F;font-size:1.4rem;z-index:2;padding-left:4px;box-shadow:0 10px 26px
rgba(0,0,0,.35)}
92     .reel-video .badge{position:absolute;top:12px;left:12px;background:rgba(12,38,31,.72);
color:var(--lime);font-size:.7rem;font-weight:700;letter-spacing:.06em;padding:.26rem
.6rem;border-radius:999px;z-index:2}
93     .reel-video
.wm{position:absolute;bottom:10px;right:12px;color:rgba(234,243,238,.78);font-size:.66rem;font-
weight:500;z-index:2}
94     .reel-body{padding:.95rem 1.05rem 1.1rem}
95     .reel-body .lbl{font-size:.72rem;font-weight:600;letter-spacing:.04em;text-
transform:uppercase;color:var(--mut)}
96     .reel-track{position:relative;height:12px;border-
radius:6px;background:#E8ECE6;margin:.55rem 0 .95rem}
97     .reel-track span{position:absolute;top:0;bottom:0;background:var(--green);border-
radius:6px}
98     .reel-stats{display:flex;gap:10px}
99     .reel-stats .m{flex:1;background:var(--paper);border-radius:10px;padding:.6rem .85rem}
100    .reel-stats .m b{display:block;font-size:1.25rem;color:var(--tx);font-weight:700}
101    .reel-stats .m em{font-size:.76rem;color:var(--mut);font-style:normal}
102    .ask{display:flex;align-items:center;gap:.55rem;margin:.85rem 1.05rem
.2rem;padding:.6rem .7rem;border:1px solid var(--line);border-
radius:12px;background:var(--paper)}
103    .ask-ic{color:var(--green-d);font-size:.95rem}
104    .ask-tx{flex:1;font-size:.88rem;color:var(--tx);font-weight:500;line-height:1.3}
105    .ask-go{width:26px;height:26px;border-
radius:8px;background:var(--green);color:#04140D;display:flex;align-items:center;justify-
content:center;font-size:.85rem;flex:none}
106    .lbl{display:flex;justify-content:space-between;align-items:center}
107    .lbl .match{color:var(--green-d);font-weight:700}
108    .reel-track span.off{background:#D7DDD4}
109    .reel-track span.on{background:var(--green)}
110    .result{display:flex;align-items:center;gap:.7rem;margin-top:.2rem;padding:.65rem
.8rem;background:var(--ink);border-radius:12px}
111    .result .play{width:34px;height:34px;border-
radius:50%;background:var(--lime);color:#0C261F;display:flex;align-items:center;justify-
content:center;flex:none;font-size:.95rem;padding-left:2px}
112    .result-tx{flex:1}
113    .result-tx b{display:block;font-size:.84rem;font-weight:700;color:#fff}
114    .result-tx em{display:block;font-style:normal;font-size:.73rem;color:var(--on-dark-
mut)}
115    .result .dl{color:var(--on-dark-mut);font-size:1.1rem}
116    .askc .q{font-weight:600;background:var(--paper);border:1px solid var(--line);border-
radius:10px;padding:.6rem .75rem;font-size:.95rem}
117    .askc .q i{color:var(--green-d);font-style:normal;margin-right:.35rem}
118    .askc .a{display:flex;gap:.5rem;color:var(--mut);font-size:.93rem;margin-
top:.75rem;padding-left:.1rem}
119    .askc .a i{color:var(--green-d);font-style:normal;margin-top:.1rem}
120    .ask-note{margin-top:1.5rem;font-size:.95rem;color:var(--mut)}
121
122    /* sections */
123    section{padding:72px 0}
124    .sec-head{max-width:60ch}

```

```

125     .sec-head .kicker{font-size:.82rem;font-weight:700;letter-spacing:.08em;text-
transform:uppercase;color:var(--green-d)}
126     .sec-head h2{margin:.5rem 0 0}
127     .sec-head p{color:var(--mut);font-size:1.08rem;margin-top:.7rem}
128
129     .grid{display:grid;gap:18px;margin-top:2.4rem}
130     .g4{grid-template-columns:repeat(4,1fr)}
131     .g3{grid-template-columns:repeat(3,1fr)}
132     .g2{grid-template-columns:repeat(2,1fr)}
133     .card{background:var(--card);border:1px solid var(--line);border-
radius:var(--r);padding:1.4rem}
134     .card .ic{width:42px;height:42px;border-radius:11px;display:flex;align-
items:center;justify-content:center;
135         background:rgba(25,179,107,.12);color:var(--green-d);font-size:1.3rem;margin-
bottom:.9rem}
136     .card h3{font-size:1.05rem}
137     .card p{color:var(--mut);font-size:.96rem;margin:.4rem 0 0}
138
139     .split{display:grid;grid-template-columns:1.1fr .9fr;gap:42px;align-items:center}
140     .panel{background:var(--ink);color:var(--on-dark);border-radius:18px;padding:2.2rem}
141     .panel h2{color:#fff}
142     .panel p{color:var(--on-dark-mut)}
143     .panel .pill{display:inline-
block;background:rgba(199,242,176,.14);color:var(--lime);font-size:.8rem;font-weight:600;
144         padding:.3rem .7rem;border-radius:999px;margin-bottom:1rem}
145
146     .road{display:grid;grid-template-columns:repeat(5,1fr);gap:14px;margin-top:2.2rem}
147     .road .step{background:var(--card);border:1px solid var(--line);border-
radius:var(--rs);padding:1.1rem}
148     .road .step .n{font-size:1.4rem;color:var(--green-d)}
149     .road .step h3{font-size:1rem;margin-top:.5rem}
150     .road .step p{font-size:.88rem;color:var(--mut);margin-top:.3rem}
151
152     .honest{background:#fff;border:1px solid var(--line);border-left:4px solid
var(--green);border-radius:0 var(--rs) var(--rs) 0;padding:1.2rem 1.4rem;margin-top:1.6rem}
153     .honest b{color:var(--tx)}
154     .honest p{color:var(--mut);margin:.3rem 0 0}
155
156     .notes{list-style:none;padding:0;margin:2rem 0 0;display:grid;grid-template-
columns:repeat(2,1fr);gap:12px}
157     .notes li{display:flex;gap:.6rem;background:var(--card);border:1px solid
var(--line);border-radius:var(--rs);padding:.9rem 1.1rem;color:var(--mut);font-size:.95rem}
158     .notes li i{color:var(--green-d);margin-top:.15rem}
159
160     details{background:var(--card);border:1px solid var(--line);border-
radius:var(--rs);padding:0 1.2rem;margin-top:10px}
161     details summary{cursor:pointer;font-weight:600;padding:1rem 0;list-
style:none;display:flex;justify-content:space-between;align-items:center;gap:1rem}
162     details summary::-webkit-details-marker{display:none}
163     details summary::after{content:"+";color:var(--green-d);font-size:1.3rem;font-
weight:400}
164     details[open] summary::after{content:"?"}
165     details p{color:var(--mut);margin:0 0 1.1rem}
166
167     .cta{background:linear-gradient(160deg,var(--ink) 0%,var(--ink2) 100%);color:var(--on-
dark);border-radius:20px;padding:3rem 2.4rem;text-align:center;margin:0 0 0}
168     .cta h2{color:#fff;max-width:20ch;margin:0 auto}
169     .cta p{color:var(--on-dark-mut);max-width:46ch;margin:.8rem auto 0}
170     .cta .hero-cta{justify-content:center}
171     .af{max-width:560px;margin:2.2rem auto 0;background:var(--card);border:1px solid
var(--line);border-radius:var(--r);padding:1.6rem 1.5rem;text-align:left}
172     .af .row{margin-bottom:1.05rem}
173     .af label{display:block;font-size:.9rem;font-weight:600;color:var(--tx);margin-
bottom:.4rem}
174     .af .hint{font-weight:400;color:var(--mut);font-size:.84rem}

```

```

175     .af input[type=text],.af input[type=email],.af select,.af
textarea{width:100%;font:inherit;font-
size:.95rem;color:var(--tx);background:var(--paper);border:1px solid var(--line);border-
radius:10px;padding:.6rem .75rem}
176     .af textarea{min-height:84px;resize:vertical}
177     .af input:focus,.af select:focus,.af textarea:focus{outline:none;border-
color:var(--green);box-shadow:0 0 0 3px rgba(25,179,107,.15)}
178     .seg{display:flex;gap:10px}
179     .seg label{flex:1;margin:0;border:1px solid var(--line);border-
radius:10px;padding:.7rem .8rem;cursor:pointer;font-weight:500;text-align:center}
180     .seg input{position:absolute;opacity:0}
181     .seg label:hover{border-color:var(--green)}
182     .seg label:has(input:checked){border-
color:var(--green);background:rgba(25,179,107,.07);color:var(--green-d);font-weight:700}
183     .opts{display:flex;flex-wrap:wrap;gap:8px}
184     .opts label{margin:0;display:inline-flex;align-items:center;gap:.4rem;border:1px solid
var(--line);border-radius:999px;padding:.4rem .8rem;font-weight:500;font-
size:.88rem;cursor:pointer}
185     .opts label:has(:checked){border-
color:var(--green);background:rgba(25,179,107,.07);color:var(--green-d)}
186     .opts input{accent-color:var(--green)}
187     .af .consent{display:flex;align-items:flex-start;gap:.5rem}
188     .af .consent input{margin-top:.2rem;accent-color:var(--green)}
189     .af .acad{display:none}
190     .af.is-academy .acad{display:block}
191     .hp{position:absolute;left:-9999px;width:1px;height:1px;overflow:hidden}
192     .form-done{max-width:560px;margin:2rem auto 0;text-
align:center;background:rgba(25,179,107,.08);border:1px solid
rgba(25,179,107,.3);color:var(--green-d);border-radius:var(--r);padding:1.4rem;font-weight:600}
193
194     footer{border-top:1px solid var(--line);padding:42px 0;color:var(--mut);font-
size:.9rem}
195     .foot{display:flex;flex-wrap:wrap;justify-content:space-between;gap:1.2rem;align-
items:center}
196     .foot a{color:var(--green-d);text-decoration:none}
197     .foot a:hover{text-decoration:underline}
198     .stamp{font-size:.82rem;color:var(--mut)}
199
200     @media (max-width:860px){
201         .nav-links{display:none}
202         .g4{grid-template-columns:repeat(2,1fr)}
203         .g3,.g2{grid-template-columns:1fr}
204         .split{grid-template-columns:1fr;gap:24px}
205         .road{grid-template-columns:repeat(2,1fr)}
206         .notes{grid-template-columns:1fr}
207         .shuttle{opacity:.1}
208         .hero-grid{grid-template-columns:1fr;gap:30px}
209         .reel-card{max-width:460px}
210     }
211 </style>
212 </head>
213 <body>
214
215 <header>
216     <div class="wrap nav">
217         <a class="brand" href="#top">Khel<span class="play" aria-
hidden="true"></span>Sutra</a>
218         <nav class="nav-links" aria-label="Primary">
219             <a href="#how">How it works</a>
220             <a href="#diff">Why it's hard</a>
221             <a href="#roadmap">Roadmap</a>
222             <a href="#faq">FAQ</a>
223         </nav>
224         <a class="btn btn-dark" href="#join">Request access</a>
225     </div>

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226     </header>
227
228     <a id="top"></a>
229     <section class="hero" style="padding:0">
230         <div class="wrap hero-grid">
231             <div class="hero-left">
232                 <span class="eyebrow">Invite-only alpha · Bengaluru</span>
233                 <h1>Every rally, indexed. <span class="hl">The dead time, gone.</span></h1>
234                 <p class="lead">KhelSutra finds every rally in your session and strips out the
knock-ups, walks and gaps ? so a full match becomes a clean highlight reel you can watch and
download. No editing, no scrubbing.</p>
235                 <div class="hero-cta">
236                     <a class="btn btn-primary" href="#join">Request alpha access</a>
237                     <a class="btn btn-ghost" href="#how">See how it works</a>
238                 </div>
239                 <p class="hero-note">A phone or GoPro on a tripod is all you need. No special
hardware.</p>
240                 <div class="loop">
241                     <span class="chip"><b>1.</b> Auto-index rallies</span>
242                     <span class="arrow">?</span>
243                     <span class="chip"><b>2.</b> Dead time removed</span>
244                     <span class="arrow">?</span>
245                     <span class="chip"><b>3.</b> Download your reel</span>
246                 </div>
247             </div>
248             <div class="hero-visual">
249                 <div class="reel-card">
250                     <div class="reel-head">
251                         <span class="dots"><i></i><i></i><i></i></span>
252                         <span class="file">match_2026.mp4 · indexed</span>
253                     </div>
254                     <div class="reel-body">
255                         <div class="lbl"><span>Rallies detected</span><span class="match">44
found</span></div>
256                         <div class="reel-track">
257                             <span class="on" style="left:2%;width:8%"></span>
258                             <span class="on" style="left:13%;width:6%"></span>
259                             <span class="on" style="left:22%;width:12%"></span>
260                             <span class="on" style="left:37%;width:7%"></span>
261                             <span class="on" style="left:47%;width:11%"></span>
262                             <span class="on" style="left:61%;width:6%"></span>
263                             <span class="on" style="left:70%;width:13%"></span>
264                             <span class="on" style="left:86%;width:9%"></span>
265                         </div>
266                         <div class="result">
267                             <span class="play" aria-hidden="true">?</span>
268                             <span class="result-tx"><b>Highlight reel · 44 rallies · 18 min</b><em>dead
time removed · ready to download</em></span>
269                             <span class="dl" aria-hidden="true">?</span>
270                         </div>
271                     </div>
272                 </div>
273             </div>
274         </div>
275     </section>
276
277     <section id="how">
278         <div class="wrap">
279             <div class="sec-head">
280                 <div class="kicker">What it does today</div>
281                 <h2>A full match, turned into rallies you can search</h2>
282                 <p>Upload a whole session or match. KhelSutra does the rest ? no editing skill
needed.</p>
283             </div>
284             <div class="grid g4">

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285         <div class="card">
286             <div class="ic" aria-hidden="true">?</div>
287             <h3>Every rally, indexed</h3>
288             <p>It finds each rally ? start, end and length ? and turns the whole match into
a searchable list. Long full-session videos are fine.</p>
289         </div>
290         <div class="card">
291             <div class="ic" aria-hidden="true">?</div>
292             <h3>Dead time removed</h3>
293             <p>Knock-ups, walks, picking up the shuttle, the gaps between points ? stripped
out. What's left is only real play.</p>
294         </div>
295         <div class="card">
296             <div class="ic" aria-hidden="true">?</div>
297             <h3>Process once, reuse</h3>
298             <p>Come back later to watch or download your reel. Re-running a video replaces
its earlier result rather than piling up copies.</p>
299         </div>
300         <div class="card">
301             <div class="ic" aria-hidden="true">?</div>
302             <h3>Stitched reel + summary</h3>
303             <p>Your selection becomes one MP4 to download and share ? with a play-time
summary and a small ?Generated by Khelsutra.guru? mark.</p>
304         </div>
305     </div>
306 </div>
307 </section>
308
309 <section id="diff" style="padding-top:0">
310     <div class="wrap">
311         <div class="split">
312             <div>
313                 <div class="kicker" style="font-size:.82rem;font-weight:700;letter-
spacing:.08em;text-transform:uppercase;color:var(--green-d)">The hard problem we crack</div>
314                 <h2 style="margin-top:.5rem">Most academy halls run several courts at once</h2>
315                 <p style="color:var(--mut);font-size:1.05rem">Academies already record a lot of
video ? and then almost never watch it, because cutting it into useful clips by hand is tedious.
The good rallies stay buried.</p>
316                 <p style="color:var(--mut);font-size:1.05rem">A single recording of a busy,
multi-court hall is exactly where ordinary video tools choke. Pulling the <b
style="color:var(--tx)">real</b> rallies out of busy, multi-court footage is the hard problem
KhelSutra is built for. It already works very well on clean, single-court recordings ? and the
multi-court case is what we're sharpening now.</p>
317             </div>
318             <div class="panel">
319                 <span class="pill">Capture tip</span>
320                 <h2 style="font-size:1.4rem">Best results: a steady, full-court shot</h2>
321                 <p>Camera behind the court, slightly elevated, whole court in frame. Keep it
still ? a fixed tripod beats handheld every time. Shoot landscape, record the whole session,
upload as-is. KhelSutra does the cutting.</p>
322             </div>
323         </div>
324     </div>
325 </section>
326
327 <section id="sharper" style="background:#EFF4EC">
328     <div class="wrap">
329         <div class="sec-head">
330             <div class="kicker">What makes it different</div>
331             <h2>It gets sharper from real academy footage</h2>
332             <p>The footage real academies contribute ? especially the busy multi-court halls ?
is what makes KhelSutra better over time. Early users aren't just customers; they're co-
builders.</p>
333         </div>
334         <div class="grid g3">

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335         <div class="card">
336             <div class="ic" aria-hidden="true">?</div>
337             <h3>Improves with the library</h3>
338             <p>When you choose to share footage and coaches confirm what it found, that
growing labelled library is what sharpens future versions ? rather than staying frozen.</p>
339         </div>
340         <div class="card">
341             <div class="ic" aria-hidden="true">?</div>
342             <h3>Your data stays yours</h3>
343             <p>We never use your video to improve the tool without your clear, explicit opt-
in. We never publish your footage. You decide what, if anything, you share.</p>
344         </div>
345         <div class="card">
346             <div class="ic" aria-hidden="true">?</div>
347             <h3>Minors are protected</h3>
348             <p>Footage involving minors is never used to improve the tool and is never
published ? full stop.</p>
349         </div>
350     </div>
351     <div class="honest">
352         <b>Read this honestly:</b>
353         <p>Today the tool gives you rally detection, reels, and summaries. It does not
retrain itself as you use it, and there's no per-person tracking. "Gets sharper over time" is
the model we're building toward, powered by contributed footage ? not a switch that's already
on.</p>
354     </div>
355 </div>
356 </section>
357
358 <section id="ask">
359     <div class="wrap">
360         <div class="sec-head">
361             <div class="kicker">Coming next to the alpha</div>
362             <h2>Soon: ask for the exact reel you want</h2>
363             <p>Your match is already a clean index of rallies ? so the natural next step is to
let you build any reel just by describing it, in plain English. This is rolling out to alpha
users.</p>
364         </div>
365         <div class="grid g3 ask-grid">
366             <div class="card askc">
367                 <div class="q"><i>?</i>Every rally over 10 seconds?</div>
368                 <div class="a"><i>?</i> 6 rallies · a 1:32 reel of the long, end-to-end
battles</div>
369             </div>
370             <div class="card askc">
371                 <div class="q"><i>?</i>Just the third game, longest first?</div>
372                 <div class="a"><i>?</i> 12 rallies, reordered by length ? that game only</div>
373             </div>
374             <div class="card askc">
375                 <div class="q"><i>?</i>The 8 best rallies from today?</div>
376                 <div class="a"><i>?</i> an 8-clip highlight from your latest session</div>
377             </div>
378         </div>
379         <p class="ask-note"><b style="color:var(--tx)">Rolling out now.</b> Targeted asks ?
by rally length, count, order, or which game ? come first; shot-type and player queries follow
as the deeper analysis arrives.</p>
380     </div>
381 </section>
382
383 <section id="roadmap">
384     <div class="wrap">
385         <div class="sec-head">
386             <div class="kicker">Where it's going</div>
387             <h2>Toward an AI assistant coach</h2>
388             <p>Fed by the growing video library, KhelSutra is heading from "finds rallies" to

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a genuine training companion. This is the direction ? not today's buttons.</p>
389     </div>
390     <div class="road">
391         <div class="step"><div class="n">?</div><h3>Shot detection</h3><p>Smashes, drops,
net shots, serves and clears within each rally.</p></div>
392         <div class="step"><div class="n">?</div><h3>Deeper analysis</h3><p>Per-rally and
per-match patterns ? what's working and what isn't.</p></div>
393         <div class="step"><div class="n">?</div><h3>Per-match history</h3><p>Track
progress over weeks and months, not just one clip.</p></div>
394         <div class="step"><div class="n">?</div><h3>Assistant coach</h3><p>Surfaces what
to work on next ? supports the real coach, never replaces.</p></div>
395         <div class="step"><div class="n">?</div><h3>More sports & languages</h3><p>Table
tennis next; Kannada and Hindi alongside English.</p></div>
396     </div>
397 </div>
398 </section>
399
400 <section id="not" style="padding-top:0">
401     <div class="wrap">
402         <div class="sec-head">
403             <div class="kicker">So expectations stay honest</div>
404             <h2>What KhelSutra is not</h2>
405         </div>
406         <ul class="nots">
407             <li><i aria-hidden="true">?</i><span>An early, invite-only trial ? useful, but
still improving. You'll hit rough edges.</span></li>
408             <li><i aria-hidden="true">?</i><span>Not a line-calling, scoring, or umpiring
system.</span></li>
409             <li><i aria-hidden="true">?</i><span>Not player tracking or
identification.</span></li>
410             <li><i aria-hidden="true">?</i><span>Not a replacement for a coach ? a tool that
saves time and adds insight.</span></li>
411             <li><i aria-hidden="true">?</i><span>Shot detection and deeper analytics are on
the roadmap, not available today.</span></li>
412             <li><i aria-hidden="true">?</i><span>Pricing isn't set ? today's invited use is
about getting it right.</span></li>
413         </ul>
414     </div>
415 </section>
416
417 <section id="faq" style="background:#EFF4EC">
418     <div class="wrap">
419         <div class="sec-head">
420             <div class="kicker">Quick answers</div>
421             <h2>Common questions</h2>
422         </div>
423         <div style="margin-top:1.6rem">
424             <details><summary>How accurate is it?</summary><p>Very good on a clean, single-
court recording. A busy multi-court hall is harder ? and is exactly what we're improving. The
best answer is to show you a sample reel of your own court rather than quote a
number.</p></details>
425             <details><summary>Can I choose which rallies go in the reel?</summary><p>Today
your reel includes the rallies KhelSutra found, with the dead time removed. Choosing by length,
count, order or which game ? just by asking in plain English ? is rolling out to alpha users
next; shot-type and player queries follow as the deeper analysis arrives.</p></details>
426             <details><summary>Do I need a special camera or hardware?</summary><p>No. A phone
or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page capture
checklist at setup.</p></details>
427             <details><summary>Is my video private? What happens to it?</summary><p>We treat
your footage as private: we never publish it, and we never use it to improve the tool without
your explicit opt-in. These are large videos that need real processing, so your video is handled
securely on our infrastructure to make your reel. Footage involving minors is never used and
never published.</p></details>
428             <details><summary>Does it work for doubles and tournaments?</summary><p>It finds
rallies in doubles too. Tournament footage varies ? a clean, fixed, full-court recording works
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well; crowded multi-court tournament halls are the hard case we're specifically working
on.</p></details>
429     <details><summary>Which sports?</summary><p>Badminton today. Table tennis and
other racket sports are on the roadmap.</p></details>
430     <details><summary>How do I join?</summary><p>It's an invite-only private trial.
The best next step is to let us record a session and show you your own court and players as a
reel. Email <a href="mailto:hello@khelsutra.guru">hello@khelsutra.guru</a>.</p></details>
431     </div>
432 </div>
433 </section>
434
435 <section id="join" style="padding-bottom:90px">
436 <div class="wrap">
437 <div class="sec-head" style="text-align:center;max-width:none">
438 <div class="kicker">Request access</div>
439 <h2>Get into the KhelSutra alpha</h2>
440 <p>Invite-only. Tell us a little about your setup ? it helps us get you the best
results, and shapes what we build next.</p>
441 </div>
442
443 <form id="af" class="af" novalidate>
444 <div class="row">
445 <label>I'm a?</label>
446 <div class="seg">
447 <label><input type="radio" name="segment" value="academy"
required><span>Coaching academy / club</span></label>
448 <label><input type="radio" name="segment" value="individual"><span>Individual
player</span></label>
449 </div>
450 </div>
451
452 <div class="row">
453 <label for="af-name">Name</label>
454 <input id="af-name" name="name" type="text" autocomplete="name" required>
455 </div>
456 <div class="row">
457 <label for="af-email">Email <span class="hint">(so we can reach
you)</span></label>
458 <input id="af-email" name="email" type="email" autocomplete="email" required>
459 </div>
460 <div class="row">
461 <label for="af-city">City <span class="hint">(optional)</span></label>
462 <input id="af-city" name="city" type="text">
463 </div>
464
465 <div class="row acad">
466 <label for="af-org">Academy / club name <span
class="hint">(optional)</span></label>
467 <input id="af-org" name="org_name" type="text">
468 </div>
469 <div class="row acad">
470 <label for="af-courts">How many courts do you record? <span
class="hint">(optional)</span></label>
471 <select id="af-courts" name="courts">
472 <option value="">Select?</option>
473 <option value="1">1 court</option>
474 <option value="2-4">2?4 courts</option>
475 <option value="5+">5+ courts</option>
476 </select>
477 </div>
478
479 <div class="row">
480 <label>Do you have a GPU machine that could process video locally?</label>
481 <div class="opts">
482 <label><input type="radio" name="has_gpu" value="yes">Yes ? a gaming / GPU PC

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or laptop</label>
483         <label><input type="radio" name="has_gpu" value="unsure">Not sure</label>
484         <label><input type="radio" name="has_gpu" value="no">No</label>
485         <label><input type="radio" name="has_gpu" value="prefer-cloud">I'd prefer it
runs in the cloud</label>
486     </div>
487 </div>
488
489     <div class="row">
490         <label>Where do your recordings live / how would you share them? <span
class="hint">(pick any)</span></label>
491         <div class="opts">
492             <label><input type="checkbox" name="storage" value="gdrive">Google
Drive</label>
493             <label><input type="checkbox" name="storage" value="onedrive">OneDrive / MS
365</label>
494             <label><input type="checkbox" name="storage" value="dropbox">Dropbox</label>
495             <label><input type="checkbox" name="storage" value="bucket">Cloud bucket
(S3/GCS)</label>
496             <label><input type="checkbox" name="storage" value="local">SD card / local
drive</label>
497             <label><input type="checkbox" name="storage" value="unsure">Not sure</label>
498         </div>
499     </div>
500
501     <div class="row">
502         <label for="af-msg">Anything else? <span class="hint">(how you record, what
you'd want?)</span></label>
503         <textarea id="af-msg" name="message"></textarea>
504     </div>
505
506     <div class="row consent">
507         <input id="af-consent" name="consent" type="checkbox" value="yes">
508         <label for="af-consent" style="margin:0;font-weight:400;color:var(--mut)">Okay
to email me about the KhelSutra alpha.</label>
509     </div>
510
511     <input class="hp" type="text" name="company" tabindex="-1" autocomplete="off"
aria-hidden="true">
512
513     <button class="btn btn-primary" type="submit" style="width:100%;justify-
content:center">Request access</button>
514     <p class="hint" style="text-align:center;margin:.85rem 0 0">We never publish your
footage; footage involving minors is never used to improve the tool.</p>
515 </form>
516
517     <p id="af-done" class="form-done" hidden>Thanks ? we've got it. We'll be in touch at
the email you gave. ?</p>
518 </div>
519 </section>
520
521 <footer>
522     <div class="wrap foot">
523         <div>
524             <div class="brand" style="font-size:1.05rem">Khel<span class="play" aria-
hidden="true"></span>Sutra</div>
525             <div class="stamp" style="margin-top:.4rem">Automatic badminton highlights .
Bengaluru, India</div>
526         </div>
527         <div style="text-align:right">
528             <div><a href="/capture.html">Capture checklist</a> . <a
href="mailto:hello@khelsutra.guru">hello@khelsutra.guru</a> . <a
href="https://khelsutra.guru">khelsutra.guru</a></div>
529             <div class="stamp" style="margin-top:.4rem">© 2026 KhelSutra . Invite-only alpha .
Reels carry a "Generated by Khelsutra.guru" mark</div>

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530     </div>
531 </div>
532 </footer>
533
534 <!-- Cloudflare Web Analytics: paste your token from Cloudflare dashboard ? Analytics ?
Web Analytics ? Add a site.
535     (Or skip this snippet entirely and enable "Automatic setup" once the domain is
proxied through Cloudflare.) -->
536 <script defer src="https://static.cloudflareinsights.com/beacon.min.js" data-cf-
beacon="{\"token\": \"REPLACE_WITH_CF_TOKEN\"}></script>
537
538 <script>
539 (function(){
540     var f = document.getElementById('af');
541     if(!f) return;
542     f.addEventListener('change', function(e){
543         if(e.target.name === 'segment') f.classList.toggle('is-academy', e.target.value ===
'academy');
544     });
545     function mailtoFallback(p){
546         var lines = [
547             'Segment: ' + (p.segment||''),
548             'Name: ' + (p.name||''),
549             'Email: ' + (p.email||''),
550             'City: ' + (p.city||''),
551             p.segment === 'academy' ? 'Academy: ' + (p.org_name||'') + ' (' + (p.courts||'')
+ ' courts)' : null,
552             'GPU: ' + (p.has_gpu||''),
553             'Storage: ' + ((p.storage||[]).join(', ')),
554             'Message: ' + (p.message||''),
555             'Consent: ' + (p.consent ? 'yes' : 'no')
556         ].filter(Boolean).join('\n');
557         return 'mailto:hello@khelsutra.guru?subject=' + encodeURIComponent('KhelSutra alpha
access') +
558             '&body=' + encodeURIComponent('Hi KhelSutra team, I would like to try the
alpha.\n\n' + lines);
559     }
560     f.addEventListener('submit', async function(e){
561         e.preventDefault();
562         if(f.company && f.company.value) return; // honeypot: silently drop bots
563         if(!f.checkValidity()){ f.reportValidity(); return; }
564         var fd = new FormData(f);
565         var p = {
566             segment: fd.get('segment'), name: fd.get('name'), email: fd.get('email'),
567             city: fd.get('city'), org_name: fd.get('org_name'), courts: fd.get('courts'),
568             has_gpu: fd.get('has_gpu'), storage: fd.getAll('storage'),
569             message: fd.get('message'), consent: !!fd.get('consent')
570         };
571         var ok = false;
572         try {
573             var r = await fetch('/api/request-access', {method:'POST', headers: {'content-
type': 'application/json'}, body: JSON.stringify(p)});
574             ok = r.ok;
575         } catch(_){ ok = false; }
576         if(ok){ f.hidden = true; document.getElementById('af-done').hidden = false; }
577         else { window.location.href = mailtoFallback(p); } // backend not live yet ?
prefilled email
578     });
579 })();
580 </script>
581
582 </body>
583 </html>
584

```

## 8. assistant (2026-06-20T10:03:12.506Z)

Now let me read the SPA console files and the engine docs in parallel.

## 9. user (2026-06-20T10:03:13.464Z)

```
1      export default function App() {
2        return (
3          <main className="min-h-screen bg-paper text-ink">
4            <header className="bg-ink text-white">
5              <div className="mx-auto flex max-w-5xl items-center gap-1 px-6 py-4">
6                <span className="text-xl font-extrabold tracking-tight">
7                  Khel<span className="text-green">?</span>Sutra{" "}
8                  <span className="font-semibold text-lime">Studio</span>
9                </span>
10             </div>
11           </header>
12
13           <section className="mx-auto max-w-5xl px-6 py-16">
14             <span className="inline-flex rounded-full bg-green/15 px-3 py-1 text-xs font-
semibold text-green">
15               operator console · scaffold (M1)
16             </span>
17             <h1 className="mt-4 text-3xl font-extrabold tracking-tight">
18               Drive the rally indexer from the browser
19             </h1>
20             <p className="mt-3 max-w-2xl text-ink/70">
21               The web SPA that runs the loop: <strong>upload ? index ? pick rallies ? reel ?
download</strong>.
22               This is the scaffold; the screens land next (see
<code>docs/PER_RALLY_SELECTION.md</code> and{" "}
23               <code>docs/LOCAL_APPLIANCE_ARCHITECTURE.md</code>).
24             </p>
25             <ul className="mt-6 space-y-1 text-sm text-ink/70">
26               <li>? Typed engine API client (<code>src/api/client.ts</code>) ? query ·
compile · highlights · config</li>
27               <li>? Dev proxy <code>/api ? :8000</code> (same-origin in the appliance)</li>
28               <li>? Structured client logging with correlation IDs
(<code>src/lib/log.ts</code>)</li>
29             </ul>
30           </section>
31         </main>
32       );
33     }
34
```

## 10. user (2026-06-20T10:03:14.387Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

## 11. user (2026-06-20T10:03:14.548Z)

```
1      # Interactive per-rally selection ? reel builder (RallyPicker)
2
3      > **Status:** `IN PROGRESS` ? design **APPROVED by owner 2026-06-20**; building the MVP
(P1?P5). Downstream milestones (multi-video auth, per-rally preview endpoint, any `/api/compile`
default change) remain owner-gated. · **Owner:** Avi · **Created:** 2026-06-20 · **Last
updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** second major tracked work item in the `khelsutra` repo (peer of
```

```

`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
7     > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9     ---
10
11     ## 0. TL;DR
12     Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally
**list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest
first") pre-selects into the **same** selection set; a sticky footer compiles the selection into
one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ?
`POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396,
307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most
important caveat: queries and chips are scoped to **length / count / order / shot-count only**;
shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be
faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the
**final compiled reel** can stream ? so MVP selects on text metadata, and visual preview/timeline
is a deliberate later milestone.
13
14     ## 1. Problem & goal
15     Today the alpha can detect rallies and stitch a reel, but reel-building is either a
**silent `stitching.min_shot_count` threshold** `[verified backend/main.py:362-383]` or a static
checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies`
while the API returns `play_segments`, so it renders nothing `[verified
backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-
video selection, not a fixed threshold**: a coach should say "give me the long rallies from this
session" and get exactly those, then fine-tune by hand.
16
17     **The concrete outcome ("good"): A coach uploads a 40-minute session; the engine finds
60 rallies. They type **"top 15 longest"**, eyeball the auto-selected cards, untick two warm-up
rallies, name it **"Tuesday ? long points"**, and download a ~6-minute reel ? without learning any
threshold knob.
18
19     **Goal: the smallest **honest** surface that turns engine-detected rallies into a user-
curated, downloadable reel, with both **fast** (query) and **precise** (checkbox) selection
sharing **one** state.
20
21     **Non-goals (alpha):** no conversational LLM (`UI_PROPOSAL.md:66` defers it); no shot-
type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt
streaming; no multi-tenant auth (single-box alpha).
22
23     ## 2. Decisions locked
24     | # | Decision | Source / date | Implication |
25     |---|-----|-----|-----|
26     | D1 | Dominant surface = selectable rally **list/grid** | Synthesis 2026-06-20 (Grid
scored 15, fit-to-data 4) | Scales to long matches via bulk+filter; shows honest fields only |
27     | D2 | Fast entry = **deterministic** client-side `parseQuery()` (no LLM) | `[verified
UI_PROPOSAL.md:66]` conversational QA deferred | Predictable, unit-testable, offline, zero API
cost |
28     | D3 | **One** shared `Set<segment_id>`, mutated by query + checkbox + (future) timeline
| Synthesis 2026-06-20 | No second compile path can diverge from what the user sees |
29     | D4 | Compile via existing `POST /api/compile {video_id, segment_ids, output_filename}`
| `[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher
`_merge_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |
30     | D5 | Query grammar (alpha) = length, count, order, **shot_count only** | `[verified
database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out
the rest |
31     | D6 | MVP backend = **zero new endpoints** | `[verified]` reuse map (§4) | Ships the
full loop on the already-shipped contract |
32     | D7 | Replace the silent `min_shot_count` cut with **explicit user selection** | Owner

```



```

framing + `[verified main.py:362-383]` | UI *warns* about the floor; it never hides why a rally
vanished |
33     | D8 | **Default selection = ON** (curate-by-removal) | Owner 2026-06-20 | Rallies load
all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX
assumes a non-empty start |
34     | D9 | **Preview-before-select is NOT a ship-blocker** for the MVP | Owner 2026-06-20 |
MVP selects on text metadata + previews the *final compiled reel*; the source/clip-stream
endpoint (per-rally preview) is the next milestone (P9), not MVP |
35     | D10 | **MVP is job-tethered** (carry `result.video_id`); no `GET /api/videos` | Owner
2026-06-20 | Smallest honest slice; multi-video home + identity scoping is P8 (gated) |
36     | D11 | `min_shot_count`: **warn, do NOT override** in the MVP | Owner 2026-06-20 | The
UI surfaces the floor; an explicit-selection override of the compile default is deferred (would
trigger the Launch Report convention) |
37
38     ## 3. Foundation ? what the engine gives us today `[verified this session]`
39     The whole design rests on the engine's real, already-shipped contract. Read against the
engine repo (`badminton-highlight-indexer`):
40
41     - **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
42     - **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the
*only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py:331-340, database.py:427-475]`.
43     - **Building a reel = `POST /api/compile {video_id, segment_ids:[int],
output_filename}`** ? loads `videos.filepath`, filters all segments to the requested
`segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and
calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
44     - **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled
MP4. Compile returns a **server-local filepath, NOT a URL** `[verified main.py:394]`, so the SPA
constructs this download URL itself. The compiled reel is the **only streamable video that
exists today** `[verified main.py:307-328]`.
45     - **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.
46     - **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos
endpoint.
47
48     **Honest fields vs fabricated/roadmap (critical for the UI):**
49     - **Honest today** (default `default_segmenter: "gemini"`, per `config.json`
`[verified]`): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a
**string**, often "Unknown") `[verified gemini.py:470-485]`. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.
50     - **Fabricated ? do NOT surface:** the `motion` demo segmenter fills `server`/`receiver`
via `random.sample(...)`, a random `shot_count`, and random `events` (serve/smash/foul/winner)
`[verified motion.py:191-198]`. `query` returns these columns + `events`, but they are not
ground truth and must stay hidden.
51     - **Roadmap ? no column, no detector:** shot-type per shot, player identity, match/point
score, game/set grouping. None exist anywhere in the schema; do not consume them `[verified
database.py whole-schema 122-148]`.
52     - **Correction-loop columns exist but are dead:**
`candidate_segments.human_label/notes/confirmed_segment_id` are defined but no endpoint writes
them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6,
`UI_ALPHA_EXECUTION_PLAN.md`).
53
54     ## 4. Design / architecture
55
56     **Reuse map (existing ? `[verified]`, no change for MVP):** `POST /api/query` (list) ?
`POST /api/compile` (stitch) ? `GET /api/highlights/{video_id}/{filename}` (download); `GET
/api/config` (floor); `GET /api/jobs/{id}` (carry `result.video_id`). **All net-new code is the
SPA** in `kshelsutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`), plus a small

```

set of *optional* real-gap engine enablers (§4.3) ? none required for MVP.

57

58     **\*\*4.1 Components (net-new SPA)\*\***

59     - **\*\*QueryBar + `parseQuery()`\*\*** ? a *\*pure, deterministic, unit-tested\** string ?  
`{filter, sort}` compiler over `duration` / `ordinal` / `shot\_count`. Renders a **\*\*parsed-pill\*\***  
("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-  
correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM,  
no network call.

60     - **\*\*RallyList/Grid\*\*** ? one card per `play\_segment`: ordinal (by `start\_time`), `mm:ss`  
start, duration (`end?start`), and a shot-count + confidence chip **\*\*only when**  
`metadata.shot\_count` is present**\*\*** (absent on detector paths that don't write it). Checkbox per  
card; the thumbnail area is a placeholder (no endpoint yet).

61     - **\*\*useRallySelection()` hook\*\*** ? owns `selectedIds:Set<number>`, `displayOrder`,  
`apply(queryResult)`, `toggle(id)`, and bulk Select-all / Clear / Invert. Unit-tested (matches  
the exec-plan's "API client + review-state logic" bar).

62     - **\*\*SelectionFooter (sticky)\*\*** ? live "N clips / total M:SS"; a **\*\*`min\_shot\_count`**  
warning**\*\*** when a selected rally would be dropped at compile (read from `GET /api/config`); reel-  
name input; one **\*\*Build\*\*** button.

63     - **\*\*CompileResult\*\*** ? `[` preview of the *\*compiled*  
reel\* + a download link.](GET)

64     - **\*\*TypedApiClient\*\*** ? `query`, `compile`, `buildDownloadUrl`; the typed seam to the  
engine.

65

66     **\*\*4.2 Data flow\*\*** `[verified anchors inline]`

67     Gemini segmenter (process time) writes `play\_segments {id, start/end\_time, duration,  
metadata.shot\_count/confidence/detector}` `[gemini.py:461-498]`

68     ? SPA `POST /api/query {video\_id}` ? `{play\_segments:[...]}` `[main.py:331-340]`  
**\*\*fix\*\*** the static UI's `data.rallies` read ? `play\_segments` `[app.js:344]`)

69     ? normalize to `RallyCard[]` + one shared `selectedIds:Set`;  
`server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **\*\*not**  
surfaced**\*\*** `[motion.py:191-198]`

70     ? all input modes mutate the **\*\*same\*\*** `selectedIds`: (a) `parseQuery()` predicate over  
duration/ordinal/shot\_count; (b) checkbox toggle + bulk ops; (c) *\*(post-MVP)\** timeline band

71     ? footer recomputes count + ?duration **\*\*client-side\*\***, and warns for any selected rally  
with `shot\_count < stitching.min\_shot\_count` `[main.py:362-383]`

72     ? **\*\*Build\*\*** ? `POST /api/compile {video\_id, segment\_ids: [...selectedIds],  
output\_filename}` ? stitcher de-dupes via `\_merge\_overlapping` `[compiler.py:311-317]`, cuts  
`time\_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393,  
compiler.py:303]`

73     ? compile returns a **\*\*server-local filepath, not a URL\*\*** `[main.py:394]` ? SPA builds  
`GET /api/highlights/{video\_id}/{output\_filename}` to preview in `` + download  
`[main.py:307-328]`.

74

75     **\*\*4.3 Engine API additions (none for MVP; tagged against real gaps)\*\***

76     | Endpoint | Purpose | Status |

77     |---|---|---|

78     | `POST /api/query` | List rallies (only path) ? fix SPA to read `play\_segments` |  
**\*\*exists\*\*** |

79     | `POST /api/compile` | Stitch selected `segment\_ids` ? MP4 | **\*\*exists\*\*** |

80     | `GET /api/highlights/{video\_id}/{filename}` | Stream/download compiled reel |  
**\*\*exists\*\*** |

81     | `GET /api/config` | Read `stitching.min\_shot\_count` for the warning | **\*\*exists\*\*** |

82     | `GET /api/jobs/{job\_id}` | Carry `result.video\_id` into the picker | **\*\*exists\*\*** |

83     | `POST /api/compile` ? add `download\_url` | Return `/api/highlights/...` so SPA stops  
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **\*\*modify\*\***  
|

84     | `GET /api/health` | Engine-offline banner ping the SPA is specified to use  
(`HOSTING\_PLAN.md:42`) | **\*\*new\*\*** |

85     | `GET /api/videos/{video\_id}/rallies` | RESTful/cacheable alias returning the same  
`{play\_segments}` shape | **\*\*new\*\*** |

86     | `GET /api/videos` | List a user's videos for a Match picker (only `get\_video(id)`  
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **\*\*new\*\*** |

87     | `GET /api/videos/{video\_id}/source` (Range) \*or\* `/clip?start=&end=` | Stream  
source/per-span video so a rally previews before selection ? neither exists | **\*\*new\*\*** |

88     | `GET /api/videos/{video\_id}/rallies/{segment\_id}/thumbnail` | Per-rally poster still

```

(no thumbnail column; WASB frames are transient) | **new** |
89 | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91 *** Dead-ends / do-NOT:** never surface `server`/`receiver`/`events` (motion-only
randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a
confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `Unknown`) and
`play_segments` has **no** numeric confidence column.
92
93 ## 5. Risk table
94 | Risk | Severity | Mitigation |
95 |---|---|---|
96 | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
97 | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
98 | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true
scrubber until source streams |
99 | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
100 | No auth / user scoping anywhere (CORS ``, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
101 | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
102 | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, $8) |
103
104 ## 6. Honest limits ? what this does NOT do
105 A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic
only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END**
weakness is a separate quality track, unaffected by this UI). "Conversational" here means a
**deterministic four-dimension parser**, not natural-language understanding. Nothing is
implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already
found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's
the M4 correction loop).
106
107 ## 7. Deliverables & progress tracker ? **source of truth**
108 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master`
auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
109
110 | ID | Deliverable | Depends on | Gated? | Status | PR |
111 |----|-----|-----|-----|-----|----|
112 | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? | |
113 | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | (this PR) |
114 | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | ? |
115 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | ? |
116 | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ? |
| ? |
117 | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | ? |
118 | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
119 | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
120 | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |

```

```

121 | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
122 | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
123
124 **Milestone rollout:** **M0/MVP = P1?P5** (pick?reel on existing API, zero new engine
endpoints) . **M1 = P6?P7** (RESTful + reliability polish) . **M2 = P8** (multi-video home) .
**M3 = P9** (per-rally preview/timeline) . **M4 = P10** (corpus correction loop) . **M5** = swap
`parseQuery()` for localized NL behind the same seam (gated on shot/player data landing).
125
126 ## 8. Open questions ? owner / external
127 **Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in §2:**
128 1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
129 2. ~~Is preview-before-select a ship-blocker?~~ ? **No**;; text-metadata MVP, preview is
P9 (D9).
130 3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
131 4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).
132
133 **Still open (nice-to-knows, non-blocking):**
134 5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is
explicitly deferred (`UI_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty
call.* *(Lean: avoid the word; show the parsed-pill instead.)*
135 6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for
A4? *Owner positioning call.*
136
137 ## 9. Definition of done
138 - [ ] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints.
139 - [ ] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are
shown; fabricated `server`/`receiver`/`events` are hidden.
140 - [ ] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens,
shows a visible parsed-pill, and is unit-tested.
141 - [ ] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden.
142 - [ ] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone
(or the static UI is superseded by the SPA).
143 - [ ] §7 tracker + this DoD updated; doc archived when P1?P5 land (MVP).
144
145 ## 10. References
146 **Engine code anchors `[verified this session]`:** `backend/main.py:121-123` (config),
`:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile +
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`
(play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475`
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe
+ entypoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
147 **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
148
149 ---
150 ### Changelog
151 - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid /
RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact
anchor re-verified against source before writing.
152 - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON .
preview not a ship-blocker . job-tethered MVP . warn-don't-override the floor); status `DRAFT ?
IN PROGRESS`; P0 done.
153

```

## 12. assistant (2026-06-20T10:03:18.056Z)

LOCAL\_APPLIANCE\_ARCHITECTURE.md doesn't exist at that path. Let me find the real docs and read the other relevant files. Let me also look at the rally-annotator repo and the engine's project

doc template and product overview.

### 13. user (2026-06-20T10:03:19.575Z)

No files found

### 14. user (2026-06-20T10:03:20.039Z)

No files found

### 15. user (2026-06-20T10:03:20.445Z)

```
1      # khelsutra docs index
2
3      ## Tracked work (this repo)
4      - **`PER_RALLY_SELECTION.md`** ? interactive pick-rallies ? reel builder (RallyPicker).
The alpha
5      SPA over the engine's rally index; hybrid grid + deterministic query bar on the
**existing**
6      `/api/query` + `/api/compile` contract. `DRAFT` (owner-gated on scope).
7      - **`REQUEST_ACCESS_FORM.md`** ? segmented lead-capture form + Worker/D1 backend
(shipped).
8      - **`PORTABILITY.md`** ? the Node+SQLite VM-host runbook (Cloudflare ? droplet parity).
9
10     ## Engine planning docs
11     Canonical planning docs currently live in the engine repo (`badminton-highlight-
indexer/docs`):
12     `UI_PROPOSAL.md` · `HOSTING_PLAN.md` · `UI_ALPHA_EXECUTION_PLAN.md`. They migrate here
when the
13     control plane lands (per ADR-009 in the engine repo's
`docs/archives/decisions/DECISIONS.md`).
14
```

### 16. user (2026-06-20T10:03:20.873Z)

```
1      # Next steps (live) ? KhelSutra brand / site
2
3      Prioritized "what to do next" for the **`khelsutra` product repo**. Refresh at each
checkpoint.
4
5      **Where we are (2026-06-20):** `khelsutra.guru` is **LIVE and fully shipped** ? landing
page + `/capture`
6      checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7      text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8      D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9      live). **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to
`master` ? deploys;
10     **CI runs `npm test` (20 green) on every PR**. The same form also runs on a dependency-
free **Node +
11     SQLite** host (`site/server.js`), **live + browsable on the D0 droplet ?
http://168.144.126.176:8787**
12     (systemd `khelsutra-site`). PRs #1?#8 merged. Docs: `docs/REQUEST_ACCESS_FORM.md`,
`docs/PORTABILITY.md`.
13
14     ## Next product threads (owner ideas, 2026-06-20)
15     - **Marketing video on the site** ? use Grok/Gemini image-generation to make a short
video (themed for
16     **serious enthusiasts**) walking the whole record ? index ? reel process, and **embed
it** on the
17     landing page. Design-first (storyboard + tone), then produce + embed.
18     - **Interactive per-rally selection** (big product feature) ? the engine detects
```

```

rallies; present them
19     in a UI so the user **picks which rallies go into the highlight** ? dynamic per-video,
**not a fixed
20     threshold/setting**. Pairs with the "ask for a reel" conversational builder already
teased on the
21     site. Lives in the alpha SPA (`web/`) over the rally index the engine produces
(persisted candidate
22     segments). **? Tracked design doc: `docs/PER_RALLY_SELECTION.md` (DRAFT, owner-gated
on scope).**
23     Key finding: the **MVP needs zero new engine endpoints** ? it rides the existing `POST
/api/query`
24     ? select `segment_ids` ? `POST /api/compile` ? `GET /api/highlights`. Owner decisions
pending in
25     the doc's S8 (default selection ON/OFF; is preview-before-select a ship-blocker; job-
tethered vs
26     multi-video MVP; whether the manual selection overrides the silent `min_shot_count`
floor).
27
28     ## Queued (site polish)
29     - **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
30     - **Cloudflare Web Analytics** ? paste the token into `site/public/index.html` (or
enable auto-setup).
31     - **`/capture` footer link** ? currently `/capture.html` (307?`/capture`); change to
`/capture`.
32     - **Droplet ? production** ? TLS + a subdomain (`vm.khelsutra.guru` via Caddy) if it
should be a real
33     standby; add **VM-path SMTP** so the Node host also emails (currently `noopNotify`).
34     - **DMARC** TXT for deliverability; delete any orphan `khelsutra-site` Worker; clean CF
D1 test rows
35     (`deploy-test@`, `email-test@`). Revoke the droplet SSH key
(`~/ssh/khelsutra_droplet`) when done.
36
37     ## Bigger arcs (design-first)
38     - **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the per-rally selection
UI above.
39     - **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-
prepare reels
40     (engine `PLATFORM_ARCHITECTURE.md` S3). The form's GPU/storage answers are the demand
signal.
41     - **Control plane** (`control-plane/`) at beta.
42
43     ## Convention
44     Substantial work gets a tracked doc in `docs/`; one PR per item off `origin/master`;
merge to `master`
45     auto-deploys (CI must be green).
46

```

## 17. assistant (2026-06-20T10:03:24.825Z)

The LOCAL\_APPLIANCE\_ARCHITECTURE.md doc lives in the engine repo per the docs index ("HOSTING\_PLAN.md / UI\_PROPOSAL.md" in `badminton-highlight-indexer/docs`). Let me read the rally-annotator repo (the public GitHub repo to link), the engine's PROJECT\_DOC\_TEMPLATE, the product overview, and the engine UI/hosting docs in parallel.

## 18. user (2026-06-20T10:03:32.841Z)

Already up to date.

---FILES---

```

./gitattributes
./github/workflows/ci.yml
./gitignore
./CHANGELOG.md
./docs/CSV_FORMAT.md
./docs/ENDING_REASONS.md
./LICENSE

```

```
./README.md
./test/dialog_layout.snapshot
./test/dialog_test.lua
./test/README.md
./vlc/rally_annotator.lua
---REMOTE---
origin https://github.com/avidullu/rally-annotator.git (fetch)
origin https://github.com/avidullu/rally-annotator.git (push)
```

## 19. user (2026-06-20T10:03:32.867Z)

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12       or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15       research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
```

```

43      <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45      ## 4. Design / architecture
46      <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48      ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49      <Who/what is protected vs NOT; residual risks stated plainly.>
50
51      ## 6. Honest limits ? what this does NOT do
52      <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54      ## 7. Deliverables & progress tracker  ? **source of truth**
55      Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57      | ID | Deliverable | Depends on | Gated? | Status | PR |
58      |----|-----|-----|-----|-----|----|
59      | P0 | ? | ? | No | ? | ? |
60
61      ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62      1. ?
63
64      ## 9. Definition of done
65      <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
66      - [ ] ?
67
68      ## 10. References
69      **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
**External:** `<cited sources>`.
70
71      ---
72
73      ### Changelog
74      - `<YYYY-MM-DD>` ? `<what changed>`.
75

```

## 20. user (2026-06-20T10:03:32.876Z)

```

1      # KhelSutra ? Product Overview & Early-User Guide
2
3      *What it does today, how to get the best out of it, and where it's going.*
4      *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6      ---
7
8      ## In one line
9
10     **KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Put a
phone or GoPro
11     on a tripod behind the court, record a session, and KhelSutra automatically finds the
rallies, stitches
12     a **highlight reel**, and gives you a **simple summary** (how many rallies, how much
real play time).
13     No special hardware, no editing, no manual scrubbing.
14
15     ---
16
17     ## The problem we're solving
18
19     Academies and serious players already record a lot of video ? and then almost never
watch it. Hours of

```



20       footage pile up on phones and memory cards because cutting it into useful clips by hand  
is tedious and  
21       nobody has the time. The good rallies, the moments worth reviewing, stay buried.  
22  
23       Most academy halls also run **\*\*several courts at once\*\***, so a single recording is a busy,  
noisy scene  
24       that ordinary video tools choke on. **\*\*Pulling the real rallies out of busy, multi-court  
footage is the  
25       hard problem KhelSutra is built to crack\*\*** ? it already works very well on clean,  
single-court  
26       recordings, and the multi-court case is exactly what we're sharpening now.  
27  
28       ---  
29  
30       **## Who it's for**  
31  
32       - **\*\*Coaching academies & coaches\*\*** ? turn raw session footage into review-ready clips  
and reels for  
33       teaching, without an editor.  
34       - **\*\*Serious players\*\*** ? get clips of your own rallies to review, keep, and share.  
35       - Today's focus is **\*\*badminton academies in Bengaluru\*\*** ? that's who we're building with  
first.  
36  
37       ---  
38  
39       **## What it does \*\*today\*\***  
40  
41       This is what's working right now, in the private web app invited users can try:  
42  
43       - **\*\*Upload a full session or match video\*\*** (from a phone on a tripod, or a GoPro). Long  
videos are fine.  
44       - **\*\*Automatic rally detection\*\*** ? it finds the rallies and where they start and end, on  
its own.  
45       - **\*\*Highlight reel\*\*** ? the rallies stitched into one MP4 you can watch, download, and  
share, with a small  
46       **"Generated by Khelsutra.guru"** mark.  
47       - **\*\*A simple summary\*\*** ? how many rallies were found and how much actual play time there  
was.  
48       - **\*\*Re-use & download\*\*** ? process once, come back to it later, download the reel.  
49  
50       That's the core loop: **\*\*record ? upload ? get back a reel + summary.\*\*** No editing skill  
needed.  
51  
52       > Like any automatic tool, it can occasionally miss a real rally or include one that  
isn't ? telling us  
53       > when that happens is genuinely useful feedback, not a sign it's broken (more on that  
below).  
54  
55       ---  
56  
57       **## How to use it (alpha)**  
58  
59       1. **\*\*Your KhelSutra associate helps you get access\*\*** to the private web app (it's  
invite-only; today the  
60       app is in English).  
61       2. **\*\*Upload your session video.\*\*** Large files upload in pieces and can take a while on  
Indian uplinks ?  
62       that's normal.  
63       3. **\*\*It processes in the background.\*\*** You can close the tab and come back later; you  
don't have to sit  
64       and watch it.  
65       4. **\*\*When it's done\*\***, you get the **\*\*highlight reel + summary\*\*** to watch and download.  
66       5. **\*\*Re-process anytime.\*\*** (Re-running the same video replaces its earlier result rather  
than keeping both.)  
67

```
68      ---
69
70      ## How to get the best out of it (capture matters)
71
72      KhelSutra works best on a **steady, full-court recording** ? the same way a coach would
73      set a camera to
74      review play. A few simple habits make a big difference to the results:
75
76      - **Put the camera behind the court, slightly elevated**, with the **whole court in
77      frame**.
78      - **Keep it still** ? a fixed tripod beats a handheld or panning shot every time.
79      - **Shoot landscape**, with reasonable lighting.
80      - **Record the whole session** and upload it as-is ? KhelSutra does the cutting.
81
82      Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style
83      cuts, very dark
84      halls) is the **hardest case** and where results suffer most. When in doubt: fixed
85      camera, full court,
86      don't touch it. *(Your field contact provides a one-page capture checklist at setup.)*
87
88      ---
89
90      ## The part that makes KhelSutra different: **it gets sharper from real academy
91      footage**
92
93      This is the heart of the product, and what sets it apart from a generic "AI clipper":
94
95      **The footage that real academies contribute is what makes KhelSutra better over time**
96      ? especially on
97      the busy multi-court halls that are the hard case. When you choose to share your
98      footage, and when
99      coaches confirm or correct what it found, that growing, labelled library is what we use
100     to sharpen the
101     tool. It's designed to improve in new versions as that library grows, rather than stay
102     frozen ? which is
103     why early users aren't just customers, they're co-builders.
104
105     **Your data stays yours ? and sharing is your choice.** We will **never** use your video
106     to improve the
107     tool without your **clear, explicit agreement** ? it is opt-in, never automatic.
108     **Footage involving
109     minors is never used to improve the tool and is never published.** We never publish your
110     footage; you
111     decide what, if anything, you share.
112
113     **How your video is handled (plainly):** these are large videos that need real
114     processing, so your video
115     is handled securely on our infrastructure to produce your reel. We don't publish it, and
116     we don't use it
117     to improve the tool without your opt-in.
118
119     > **Read this honestly:** today the tool gives you rally detection + reels + summaries.
120     It does **not**
121     > retrain itself as you use it, and there is no per-person tracking. The "gets sharper
122     over time" above is
123     > the model we're building toward, powered by contributed footage ? not a switch that's
124     already on.
125
126     ---
127
128     ## Where it's going ? toward an **AI assistant coach**
129
130     Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
131     library above ?
132     turns it into a genuine training companion:
```

115

116 - **\*\*Shot detection\*\*** ? recognising smashes, drops, net shots, serves, and clears within each rally.

117 - **\*\*Deeper per-rally and per-match analysis\*\*** ? patterns, tendencies, what's working and what isn't.

118 - **\*\*Per-match history\*\*** ? track progress over weeks and months, not just one clip.

119 - **\*\*Coaching insights\*\*** ? an **\*\*AI assistant coach\*\*** that supports (never replaces) the real coach,

120 surfacing what to work on next.

121 - **\*\*More sports\*\*** ? table tennis next, then other racket/court sports.

122 - **\*\*Your language\*\*** ? Kannada and Hindi, alongside English.

123

124 > **\*\*Read this honestly:\*\*** the items in this section are the **\*\*direction\*\***, not today's buttons. What you

125 > can use **\*today\*** is rally detection + highlight reels + the play-time summary. The deeper analysis arrives

126 > as the video library grows ? which is exactly why your footage and feedback matter so much.

127

128 ---

129

130 **## What feedback helps us most (for alpha users)**

131

132 You don't need to be technical ? just tell us, in your words:

133

134 - **\*\*When it missed a real rally\*\***, or **\*\*grabbed something that wasn't a rally.\*\***

135 - Whether the **\*\*reel is the right length and order\*\*** for how you'd actually use it.

136 - **\*\*What numbers or details you wish the summary showed.\*\***

137 - Your real **\*\*capture conditions\*\*** ? how many courts were running, camera/phone, lighting.

138 - Anything that would make this **\*\*genuinely useful for your academy\*\*** (or a reason it wouldn't be).

139

140 **\*(Send feedback however is easiest ? your KhelSutra associate will help you with that.)\***

141

142 ---

143

144 **## What we're focused on over the next ~4 weeks**

145

146 1. **\*\*Growing the library of real academy footage\*\*** ? especially busy **\*\*multi-court\*\*** halls (the hard,

147 valuable case). This is the single biggest lever on quality.

148 2. **\*\*Field conversations with academies\*\*** ? understanding how coaches and players actually work with

149 video, and what would make this genuinely useful.

150 3. **\*\*Recording sessions and showing academies their own footage\*\*** as reels ? **\*\*"this is what it looks**

151 **like for you."\***

152 4. **\*\*Sharpening multi-court accuracy\*\*** and inviting a few more academies into the private trial.

153

154 ---

155

156 **## What KhelSutra is **\*\*not\*\*** (so expectations stay honest)**

157

158 - It's an **\*\*early, invite-only trial\*\*** ? useful, but still improving; you'll hit rough edges.

159 - It is **\*\*not a line-calling, scoring, or umpiring system\*\***, and **\*\*not player tracking / identification\*\***.

160 - **\*\*Shot detection and the deeper "AI coach" analytics are on the roadmap, not available today.\*\***

161 - It is **\*\*not a replacement for a coach\*\*** ? it's a tool that saves coaches time and adds insight.

162 - **\*\*Pricing isn't set\*\*** ? we're still learning what's fair; today's invited use is about getting it right.

```

163
164 ---
165
166 ## Quick answers to common questions
167
168 ***How accurate is it?***
169 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is
170 exactly what
171 we're improving. The best answer is to *show* a sample reel rather than quote a number.
172
173 ***Do I need a special camera or hardware?***
174 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a
175 one-page checklist.
176
177 ***Is my video private? What happens to it?***
178 We treat your footage as private: we never publish it, and we never use it to improve
179 the tool without
180 your explicit opt-in. These are large videos that need real processing, so your video is
181 handled securely
182 on our infrastructure to make your reel. **Footage involving minors is never used to
183 improve the tool and
184 is never published.**
185
186 ***Can it tell me the score / who won / call lines?***
187 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.
188
189 ***Does it work for doubles and tournaments?***
190 It finds rallies in doubles too. Tournament footage varies: a clean, fixed, full-court
191 recording works
192 well; crowded multi-court tournament halls are the hard case we're specifically working
193 on.
194
195 ***Will it get better?***
196 Yes ? that's the whole idea. The more real footage runs through it (with permission) and
197 the more coaches
198 correct it, the sharper future versions get, and the sooner shot detection and deeper
199 analysis arrive.
200
201 ***Which sports?***
202 Badminton today; table tennis and other racket sports are on the roadmap.
203
204 ***When can I use it / how do I join?***
205 It's an invite-only private trial. If you're interested, the best next step is to let us
206 record a session
207 and show you your own court and players as a reel.
208
209 ---
210
211 *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*
212 *Field associates: the role description is in the Field Associate flyer, and the
213 conversation script and
214 visit rules are in the separate Field Visit Playbook.*
215
216

```

## 21. assistant (2026-06-20T10:03:36.829Z)

Now let me read the rally-annotator README (the public GitHub repo to link) and the engine docs referenced (UI\_PROPOSAL.md, HOSTING\_PLAN.md, OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md) to ground honesty about train-your-own-model status. Let me also check the engine's NEXT\_STEPS and the worker/train code that was just built.

## 22. user (2026-06-20T10:03:37.186Z)

```

1      # Rally Annotator
2

```

```

3      [![tests](https://github.com/avidullu/rally-
annotator/actions/workflows/ci.yml/badge.svg)](https://github.com/avidullu/rally-
annotator/actions/workflows/ci.yml)
4
5      A tiny VLC plugin to hand-label rally start/end + point-stop reason while you
watch a match,
6      for net-separated racquet sports ? badminton, tennis, table tennis, pickleball,
padel.
7
8      You watch the video in VLC, pause/scrub freely, and click Mark START / Mark END
(with a
9      reason). It writes one CSV row per rally ? clean ground-truth labels for
training/evaluating rally
10     (point) segmentation models, or just for cutting highlights.
11
12     ```
13     rally_number,start_time,end_time,ending_reason,sport,shots_count
14     1,8.800,11.500,winner,badminton,9
15     2,24.389,46.589,unforced_error,badminton,21
16     ```
17
18     Times are decimal seconds; `shots_count` is an optional rally shot/stroke count
(blank if you skip it).
19     See [docs/CSV_FORMAT.md](docs/CSV_FORMAT.md).
20
21     ## Why
22     Labeling rally boundaries is the slow, expensive prerequisite for any rally-detection
model. Most tools
23     make you transcribe timestamps by hand. This lets a rater do it inside the player they
already use,
24     pausing to the exact frame before each mark ? turning "watch a match" into "produce
golden labels."
25
26     ## Install
27     1. Copy `vlc/rally_annotator.lua` into your VLC Lua extensions folder:
28         - Windows: `%APPDATA%\vlc\lua\extensions\`
29         - macOS: `~/Library/Application Support/org.videolan.vlc/lua/extensions/`
30         - Linux: `~/.local/share/vlc/lua/extensions/`
31         (create the `extensions` folder if it doesn't exist)
32     2. In VLC: Tools ? Plugins and extensions ? Reload extensions (or just restart VLC).
33     3. Enable it from the View menu ? Rally Annotator. A small dialog opens and
stays open while you watch.
34
35     Requires VLC 3.0.x. (VLC 4.0 changed the Lua input/listener API; targeting 3.x for
now.)
36
37     ## Use
38     Playback is built in ? the Back 5s · Play / Pause · Fwd 5s row drives the VLC
player from the
39     annotation window, so you can pause, label, and resume without ever switching to the
main VLC window. Play / Pause
40     is a single toggle (it resumes when paused, pauses when playing).
41
42     1. Pick the Sport (it stays set across rallies).
43     2. When a rally begins, click Mark START ? it snapshots the current playback time
into the Start field
44         (pause/scrub first for frame accuracy; you can also edit the field by hand).
45     3. When the rally ends, click Mark END (fills the End field). This arms the
rally; nothing is written yet.
46     4. Choose the Ending reason (sits between Mark END and Save Rally; defaults
to unknown), optionally
47         type a Number of shots (the rally's shot/stroke count ? leave blank to skip),
then click
48         Save Rally to write one CSV row. The reason resets to unknown after every
save (and the shots field

```

49 clears too) ? so neither silently reuses the previous rally's value, and you can pick the same reason on

50 consecutive rallies. Use the **\*\*Next rally #\*\*** field to resume/insert numbering anywhere.

51 5. **\*\*Recent rallies\*\*** list: select a row, then **\*\*Edit selected\*\*** (loads it back into the fields to fix

52 start/end/reason/sport ? **\*\*Save changes\*\***) or **\*\*Delete selected\*\***. **\*\*Undo last\*\*** removes the most recent row ?

53 the button shows which one, e.g. ``Undo last (#7)``, and the status panel shows that row's details ? or clears an

54 in-progress mark, or cancels an edit.

55 6. **\*\*Help\*\*** button ? usage + an [ending-reason decision guide](docs/ENDING\_REASONS.md), rendered inside the dialog.

56

57 **\*\*Output:\*\*** ``<video-stem>.rallies.csv`` next to the video (falls back to your home dir if the path can't be

58 resolved). Re-opening the extension **\*\*continues\*\*** rally numbering from the existing file ? no duplicate IDs.

59

60 **\*\*Resuming a half-finished video:\*\*** labels are saved to that CSV as you go. Open the **\*\*same video\*\*** and enable the

61 extension and it **\*\*reloads your existing rallies\*\*** (shown in the Recent list) and continues numbering ? so you can

62 stop and pick up later. Easiest flow: **\*\*open the video first, then enable the extension.\*\*** If you enable it **\*\*before\*\***

63 the video is loaded, that's fine too ? the moment you click **\*\*Mark START\*\*** (the video is playing by then) the tool

64 switches to that video's own ``<video-stem>.rallies.csv`` and loads any rallies already saved for it, so nothing is

65 written to the home-dir fallback by mistake. If you switch videos with the dialog already open, click **\*\*Refresh\*\*** to

66 point at the current video and load its rallies. (Playback **\*\*position\*\*** isn't restored ? scrub to where you stopped.)

67

68 > **\*\*Where's the help / "About" text?\*\*** The empty **\*\*"About ?lua"\*\*** box under **\*\*Tools ? Plugins and extensions ?**

69 > **\*\*Add-ons\*\*** is VLC's own add-on info dialog; its body ("Lua script") is a constant baked into VLC and **\*\*can't** be

70 > set by an extension**\*\***. Use the **\*\*Help\*\*** button inside the Rally Annotator dialog instead. (The descriptor's

71 > description does show in the **\*\*Active Extensions\*\*** tab's **\*\*"More information"\*\*** dialog, and the **\*\*Active Extensions\*\***

72 > list shows the title with its version, e.g. **\*\*`Rally Annotator v1.6.3`\*\***, so you can confirm which build is loaded.)

73

74 **## Sports & taxonomy**

75 Net-separated racquet sports share a forced/unforced-error point-stop taxonomy, so one tool covers them all:

76

77 | Sport | Notes |

78 |---|---|

79 | badminton | the reference sport |

80 | tennis | service\_fault = fault/double-fault; let supported |

81 | table\_tennis | let (net serve) supported; fast cadence |

82 | pickleball | service\_fault for faults; treat "let" per local rules |

83 | padel | net-separated; rally boundaries as in tennis |

84

85 ``ending_reason`` ? ``{unknown, winner, forced_error, unforced_error, service_fault, let, other}`` ? a shared

86 vocabulary across these sports. Keep to these values for clean downstream aggregation.

87

88 **### What the ending reasons mean**

89 ``unknown`` is the **\*\*default\*\*** (the field resets to it after every save); every other

reason **\*\*except `winner`\*\*** is

90 charged to the side that **\*\*lost\*\*** the rally.

```

91
92     | Reason | When the rally ended because? |
93     |---|---|
94     | `unknown` | **default** ? not classified yet. A save with no reason picked records
`unknown` (never the previous rally's reason); set a specific reason when you can. |
95     | `winner` | the last shot landed **in** and went unreturned (opponent couldn't reach it
/ only waved). A clean ace counts here. |
96     | `forced_error` | the loser **missed** (out or into the net) while **under pressure** ?
stretched, rushed, jammed, handling the opponent's pace/spin/depth. |
97     | `unforced_error` | the loser **missed a routine shot** they had time **and** position
to make, with little or no pressure. |
98     | `service_fault` | the point ended **on the serve** ? into the net, out of the service
box, illegal action/foot fault, or a double fault. |
99     | `let` | the rally is **replayed** under the rules, no point scored (e.g. a
tennis/table-tennis serve net-cord that's otherwise good, outside interference). |
100    | `other` | none of the above ? occluded footage, injury/retirement, penalty, hindrance.
Use sparingly. |
101
102    **The cases people ask about:** a ball/shuttle landing **out** is the *hitter's* error ?
`forced_error` only if they
103    were under pressure, otherwise `unforced_error` (it's **never** automatically forced,
and never a winner for the
104    opponent). **Into the net** during a rally is the same; a **serve** into the net is
`service_fault`. When unsure
105    forced-vs-unforced, **default to `unforced_error`**. See the full decision guide with
examples and per-sport rules
106    in **[docs/ENDING_REASONS.md](docs/ENDING_REASONS.md)** (also available via the in-
plugin **Help** button).
107
108    ## Roadmap
109    - [ ] Live-test the v1.6 dialog in VLC (single Play / Pause toggle, optional Number of
shots, two-step Save,
110        `unknown`-default reason, editable times, Next rally #, Recent-rallies
Edit/Delete) across all five sports.
111    - [ ] Optional per-sport reason presets / hotkeys.
112    - [ ] Alternative front-ends for power users / remote raters: a `python-vlc` + Tk/Qt app
with global keyboard
113        shortcuts (S/E/1?6/U), and a zero-install HTML5 `` page that exports the
same CSV.
114    - [x] Optional `shots_count` column (v1.6). Further extra columns (e.g. server/receiver)
still TBD behind a toggle.
115
116    ## Tests
117    The dialog logic has a headless test suite that stubs VLC's `vlc` API (widgets +
playback), loads the real
118    extension, drives its callbacks, and asserts the results ? run with **Lua 5.1** (the
interpreter VLC 3.x embeds):
119
120    ```bash
121    lua5.1 test/dialog_test.lua          # exit 0 = pass; covers reason reset, shots column,
numbering, undo, help toggle, Play/Pause toggle
122    ```
123
124    It covers all the logic but not VLC's GUI rendering or real playback (there's no
headless VLC UI automation) ? those
125    are checked by loading in VLC with `vlc -vv --file-logging` (must scan with no `Error
loading` line) plus a manual
126    click-through. See [test/README.md](test/README.md).
127
128    ## Contributing
129    Issues and PRs welcome ? especially live test reports per sport/OS and small UX fixes.
The plugin is a single
130    self-contained Lua file (`vlc/rally_annotator.lua`) using only documented VLC 3.x Lua
APIs; please keep
131    `test/dialog_test.lua` green (`lua5.1 test/dialog_test.lua`) and add a case for new

```

```

dialog logic.
132
133     ## License
134     MIT ? see [LICENSE](LICENSE).
135

```

### 23. user (2026-06-20T10:03:37.852Z)

```

1      # Output CSV format
2
3      One row per rally. Header is written once when the file is created.
4
5      ```
6      rally_number,start_time,end_time,ending_reason,sport,shots_count
7      1,8.800,11.500,winner,badminton,9
8      2,24.389,46.589,unforced_error,badminton,21
9      3,49.183,54.683,let,badminton,
10     ```
11
12     | Column | Type | Notes |
13     |---|---|---|
14     | `rally_number` | int | 1-based, monotonic; continues across re-opening the same CSV |
15     | `start_time` | float | rally start, **decimal seconds** from the video start |
16     | `end_time` | float | rally end, decimal seconds (always `> start_time`) |
17     | `ending_reason` | enum | `unknown` (default) / `winner` / `forced_error` /
`unforced_error` / `service_fault` / `let` / `other` ? see the decision guide in
[ENDING_REASONS.md](ENDING_REASONS.md) |
18     | `sport` | enum | `badminton` / `tennis` / `table_tennis` / `pickleball` / `padel` |
19     | `shots_count` | int | **optional** ? number of shots/strokes in the rally. **Blank**
when not entered (as in row 3 above); added in v1.6. Older CSVs without this column still load.
|
20
21     Notes:
22     - **Decimal seconds**, not `mm:ss` ? downstream loaders parse floats.
23     - Rows are appended live; **Undo last** truncates the last row (one level).
24     - Consumers should read by **column name** (the header is stable but column *order* may
evolve; extra
25     columns may be added later). Most CSV `DictReader`-style loaders already tolerate
extra columns.
26
27     ## Ingesting the labels
28
29     This CSV is intentionally compatible with common rally-segmentation tooling that reads
30     `rally_number,start_time,end_time` by header name and drops rows where `end <= start`:
31
32     ```bash
33     # Generic: any tool that takes a labels CSV of start/end seconds.
34
35     # Example ? score a detector against these labels, or slice clips by them
36     # (badminton-highlight-indexer loaders read --labels <this.csv> directly):
37     python -m backend.eval.calibrate_wasb --trajectory traj.csv --labels match.rallies.csv
--fps 59.94 --frame-width 1920
38     python -m backend.tools.rally_slicer --video match.mp4 --labels match.rallies.csv
--rallies all
39
40     # Example ? register as curated golden data in sports-data-collector:
41     python -m src.cli.curate import-local --sport badminton --video-path match.mp4
--annotations-path match.rallies.csv
42     ```
43
44     The `sport` column is self-documenting; tools that take a `--sport` flag can ignore it.
45

```

### 24. user (2026-06-20T10:03:38.263Z)



```

1      # Ending reasons ? what they mean & how to label them
2
3      The `ending_reason` column records why the rally (point) ended, judged from the
4      last shot. There are seven values, shared across all the net-separated racquet
sports
5      this tool supports (badminton, tennis, table tennis, pickleball, padel):
6
7      ```
8      unknown | winner | forced_error | unforced_error | service_fault | let | other
9      ```
10
11     `unknown` is the default ? the annotator resets the reason to `unknown` after every
save, so a
12     rally you didn't classify is honestly recorded as `unknown` (never the previous rally's
reason). Set a
13     specific reason when you can; `unknown` simply marks rallies still needing a verdict.
14
15     One core principle: every reason except `winner` (and the neutral `unknown`) is
charged to the
16     player/side that LOST the rally. `winner` is the only one that credits the side that
won it.
17
18     This guide is written for a human rater clicking through a match. Where a rule comes
from
19     the laws of the sport it is marked [rule]; where it is an analytics convention (a
judgement call that charters agree to apply consistently) it is marked [convention].
20
21
22     ---
23
24     ## The decision procedure (top to bottom ? stop at the first match)
25
26     1. Was the rally replayed under the rules, with no point scored?
27         (e.g. a tennis/padel serve that clips the net and lands in the correct box; a table-
tennis
28         serve that touches the net and is otherwise good; outside interference; receiver not
ready.)
29         ? `let`. *A net-cord during open rally play that lands in is NOT a let ? keep
going.*
30
31     2. Did the point end on the SERVE because of a server fault?
32         (serve into the net, serve outside the correct service court, illegal service action
/ foot
33         fault, or a tennis/padel double fault.)
34         ? `service_fault`.
35
36     3. Did the winning side's last shot land IN and go unreturned ? the opponent
couldn't
37         reach it, or only waved at it without meaningful contact? (A clean untouched serve /
ace
38         counts here.) ? `winner`.
39
40     4. Otherwise the rally ended on a MISS by the loser (ball/shuttle out, into
the net,
41         or not legally returned). Now judge pressure:
42
43         > The pressure test: Did the loser have both enough time AND court position
to make a
44         > routine return, given the opponent's immediately preceding shot (its placement,
pace,
45         > power, spin, depth, angle)? Ask: "Would a typical competent player at this level
be
46         > expected to make that shot?" [convention]
47
48         - Yes ? had time and position, missed a routine ball ? `unforced_error`.
49         - No ? time or position was taken away; the miss was induced ?

```

```

**`forced_error`**.
50
51 5. **Can't judge it, or it ended off-court** (occluded footage, injury/retirement,
code/point
52 penalty, hindrance, equipment failure) ? **`other`**. Use sparingly.
53
54 **Tie-breaker [convention]:** when you genuinely can't tell forced from unforced,
**default to
55 `unforced_error`**. Only mark `forced_error` when you can *point to the specific
pressure* (the
56 wide/fast/deep/spinny ball that removed the player's time or position). This keeps
`forced_error`
57 a positively-evidenced label and stops it from inflating. The forced/unforced boundary
is a real
58 *continuum* ? charters disagree on borderline calls, so fix the default in writing for
your team.
59
60 ---
61
62 ## The six reasons, defined
63
64 ### `winner`
65 The point-ending shot landed **in** and the opponent did **not** make a meaningful
return
66 (couldn't reach it, or only waved). Credit to the side that won the rally. A clean
untouched
67 legal serve (an **ace**) is conventionally a `winner`, not a `service_fault`.
68 *Examples:* a smash that bounces untouched; a deceptive drop the opponent never reaches;
a
69 passing shot down the line.
70 *Not a winner:* if the opponent reaches it but mishits **under pressure**, that is
**their**
71 `forced_error`, not your `winner`.
72
73 ### `forced_error`
74 A miss by the **loser** (out, into the net, or not legally returned) where the
opponent's
75 previous shot applied enough pressure ? placement, pace, power, spin, depth, angle ?
that the
76 player did **not** have time *and* position for a routine return. They were *made* to
miss.
77 *Examples:* a wide, heavy serve return netted while lunging at full stretch; a hard
smash dug up
78 but floated long because the player was off-balance and rushed.
79
80 ### `unforced_error`
81 A miss by the **loser** on a shot they had **time AND position** to make routinely, with
little
82 or no pressure. They gave it away.
83 *Examples:* a comfortable mid-court ball drilled into the net; a relaxed clear pushed
past the
84 back line; a sitter smashed long.
85
86 ### `service_fault`
87 A fault by the **server on the serve itself** that loses the point or the serve ? judged
against
88 the sport's *service* rules, not general rally play. Includes: serve into the net, serve
out of
89 the correct service court/box, illegal service action (e.g. badminton contact above 1.15
m,
90 ITTF illegal toss, pickleball illegal motion), foot fault, and a tennis/padel **double
fault**.
91 *Note:* a **first**-serve fault followed by a good second serve is **not** charged here
? only the
92 serve that actually loses the point is. A clean ace is a `winner`, not a

```

```

`service_fault`.
93
94     ### `let`
95     A **replay** defined by the laws, where **nobody is charged** a point ? the rally
"didn't count":
96     a tennis/padel service net-cord that still lands in the correct box (replayed, unlimited
times);
97     a table-tennis serve that touches the net and is otherwise correct; an outside
interruption; the
98     receiver not ready. Use **only** when the rule says *replay*.
99     *Caution:* a net-cord during a **rally** (not the serve) that lands in is live ? **not**
a `let`.
100
101     ### `other`
102     Catch-all for endings that fit none of the above: the outcome is genuinely
unclear/occluded on
103     video; or the point ended administratively (injury retirement, code/point penalty,
hindrance
104     call, equipment failure) rather than on a struck shot. Prefer a specific category
whenever the
105     footage supports it.
106
107     ---
108
109     ## The cases people ask about most
110
111     These are the calls that trip raters up ? resolved precisely.
112
113     ### A. The ball/shuttle lands **OUT** (past the baseline or outside the sidelines)
114     This is **always the hitter's error** ? the side that **lost** the rally. The opponent
merely let
115     it sail (or it cleared the line untouched), so it is **never** the opponent's `winner`.
Decide
116     `forced` vs `unforced` purely by **pressure on the hitter** at the moment they struck it
? look at
117     the opponent's *previous* shot, not the out ball itself:
118     - Rushed, stretched, off-balance, or handling heavy pace/spin/depth ?
**`forced_error`**.
119     - Set and balanced, with time on a routine ball, and still sprayed it ?
**`unforced_error`**.
120
121     "Beyond the baseline" vs "outside the sideline" doesn't change the logic ? both are
*out*, both are
122     the hitter's miss. (In tennis and badminton the **line is in** ; this applies only when
it lands
123     fully outside.) **So "out" is not automatically `forced`** ? that label needs visible
pressure;
124     otherwise it's `unforced`.
125
126     ### B. The ball/shuttle goes **INTO the net** (fails to cross)
127     Same as an "out" ball ? it's the hitter's error, and the **pressure test** decides:
`forced_error`
128     if they were rushed/stretched, `unforced_error` if they dumped a routine ball.
129     **Exception:** if the failure-to-cross happens **on the serve**, it is a
**`service_fault`** (in
130     tennis/padel, a missed second serve = double fault), not a rally error ? the rally never
legally
131     began.
132
133     ### C. The **net-cord / net tape** (ball or shuttle clips the top)
134     - **During a rally:** it is **live and good** in all of these sports. Play continues, so
the rally
135     ends on whatever happens **next**: if the opponent can't reach the dribbler ? it's a
`winner`
136     (a "net-cord winner"); if they scramble and then miss ? judge *that* miss as

```

```

forced/unforced.
137     A rally net-cord is **never** a `let`.
138     - **On the serve ? sports differ:**
139       | Sport | Serve clips the net and? |
140       |---|---|
141       | **Tennis / Padel** | ?lands in the correct box ? **`let`** (replay). Lands outside ?
`service_fault`. |
142       | **Table tennis** (ITTF) | ?is otherwise correct ? **`let`** (serve replayed). |
143       | **Badminton** (BWF) | ?passes over and lands in the service court ? **live, play
on** (no service let). But a serve shuttle **caught/suspended on the net** is a
**`service_fault`** (BWF Law 13.2), *not* a let. |
144       | **Pickleball** (2026 USAP) | ?lands in the correct court ? **live, play on** (the
old service-let was removed). Lands out ? `service_fault`. |
145
146     ---
147
148     ## Sport-specific notes
149
150     - **Tennis (ITF):** double fault ? `service_fault`; service net-cord into the box ?
`let`; lines
151       are **in**; a clean ace ? `winner`.
152     - **Badminton (BWF):** service contact must be with the whole shuttle below **1.15 m**
from the
153       court surface (Law 9.1.6; the older "above the waist" wording is the alternative trial
law). A
154       shuttle landing outside the lines or failing to pass the net is a fault; a **serve
caught on the
155       net is a fault** (`service_fault`), while a mid-rally suspended shuttle (Law 14.2.3,
only *after*
156       the serve is returned) is a `let`. Lines are **in**.
157     - **Table tennis (ITF):** a serve touching the net assembly but otherwise correct is a
**`let`**
158       (replayed); in a rally, a ball that touches the net and still lands on the opponent's
court is a
159       **good return**.
160     - **Pickleball (USAP 2026):** **no** service let ? a net-clip that lands in the correct
court is
161       live; illegal serve motion / net-clip-out ? `service_fault`.
162     - **Padel (FIP):** like tennis for the serve net-cord `let`. Wall play means a rally
ball off the
163       glass after one bounce is still live. (One edge case ? a let-bound serve that touches
the net and
164       then the side **wire mesh** before its 2nd bounce ? is finicky; verify against current
FIP text.)
165
166     ---
167
168     ## Consistency tips
169
170     - The forced/unforced split is a **judgement call on a continuum**. Pick the default
171       (`unforced` when unsure) and apply it the same way every time ? consistency matters
more than
172       any single borderline call.
173     - If you're building a rating *team* or a regression set, consider logging a confidence
note on
174       borderline forced/unforced calls so they can be audited or re-bucketed later.
175     - These six values are deliberately a small, shared vocabulary. Keep to them (don't
invent new
176       spellings) so downstream aggregation stays clean.
177
178     ### Mapping to tennis charting codes (optional)
179     If you cross-reference the [Match Charting
Project](https://github.com/JeffSackmann/tennis_MatchChartingProject):
180     `winner = *`, `forced_error = #`, `unforced_error = @` (with location tags `n` = net,
`w` = wide,

```

```

181     `d` = deep, `x` = wide+deep). These are charting conventions layered on top of the
official rules.
182
183     ---
184
185     ## Sources
186
187     - Match Charting Project ? quick-start & codes:
<https://www.tennisabstract.com/blog/2015/09/23/the-match-charting-project-quick-start-guide/>
188     - "The continuum of errors" (forced vs unforced is a continuum):
<http://www.tennisabstract.com/blog/2017/01/13/the-continuum-of-errors/>
189     - Leo Levin's pressure definition of unforced error:
<https://www.tennis.com/news/articles/what-is-an-unforced-error-a-meditation-on-the-tennis-
player-s-least-favorite-stat>
190     - ITF Rules of Tennis: <https://www.itftennis.com/en/about-us/governance/rules-and-
regulations/>
191     - BWF Laws of Badminton: <https://www.worldbadminton.com/rules/>
192     - ITTF Table Tennis rules (Law 2.7 / 2.9): <https://www.ittf.com/handbook/>
193     - USA Pickleball rulebook: <https://usapickleball.org/what-is-pickleball/official-
rules/>
194     - FIP Rules of Padel: <https://www.padelfip.com/rules-of-padel/>
195

```

## 25. assistant (2026-06-20T10:03:43.383Z)

Now I have a strong picture of the rally-annotator (the public repo to link) and the marketing/console surfaces. Let me verify the honesty-critical claim: the "train your OWN model" capability. The engine recently shipped a "worker train" stub (commit d5407eb). Let me check the engine's owned-model plan and the worker train code to be honest about real-vs-roadmap. Also check web's index.css/tailwind config for reusable brand tokens, and the site capture.html as a precedent marketing sub-page.

## 26. user (2026-06-20T10:03:47.683Z)

Already up to date.  
---OWNED-MODEL-DOC head---

## Owned-model implementation plan (roadmap) ? DRAFT for owner review

**\*\*Status:\*\*** PLAN / roadmap. No owned-model *implementation* starts until the owner greenlights a specific milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a training source).  
**\*\*Grounded in:\*\*** `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md`, the n=6 LOVO + learned prototype (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy workflow (2026-06-09).

### Design principle (owner directive): *\*envision now, build later\**

Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be *\*\*architecturally envisioned now\*\** so we never hit a mammoth refactor. We *\*\*design the seams\*\** (data contracts, export boundaries, the event store) up front and *\*\*implement behind them\*\** when each milestone is greenlit. Punting a *\*feature\** is fine; punting its *\*seam\** is not.

### North Star (reframed per PR #61 review)

- **\*\*Immediate deliverable:\*\*** a *\*\*training framework + harness\*\** that produces a model with *\*\*very high precision/ recall for extracting highlights + rallies from a clip\*\**. *\*Quality is the first-order objective.\**
- **\*\*Eventual product:\*\*** a *\*\*single, sport-agnostic, conversational\*\** video model ? upload a clip, then ask

contextual questions ("longest rally", "make a clip of all service faults", "rallies over 20 shots").

- **\*\*On-device is LATER\*\*** (a model-compression goal so power users save upload cost/hassle) ? definitely on the roadmap, but **\*\*not\*\*** the immediate target. Don't over-index the strategy on it now.
- **\*\*The moat = the consented dataset + the eval harness, not the weights.\*\***

## Hard guardrails (non-negotiable ? "never corrupted")

- **\*\*Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT / Apache-2.0 / BSD only.\*\***  
(Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral),  
**\*\*Gemma 3n\*\*** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.
- **\*\*Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.\*\*** Also forbidden: public grounding-CoT datasets that were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision (a real, binding cost).
- **\*\*Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.\*\***
- ? **\*\*Base-model pretraining provenance is unauditale for \*every\* open VLM (incl. Qwen3-VL).\*\***  
Our "no paid-LLM training" rule is enforceable at *\*our fine-tuning data\** layer, not the base's pretraining ? an **\*\*explicit owner risk-acceptance\*\*** is required to use any open VLM. (Governed by the base's license, not our pipeline.)

## The model strategy (verified 2026-06-09)

- \*\*Framework ? base\*\*** (the Gemma-4-vs-ms-swift clarification): a *\*framework\** is the forge (runs SFT/GRPO, emits weights); a *\*base\** is the metal you forge. You pick one framework and feed it any clean base.
- **\*\*Framework: ms-swift (Apache-2.0)\*\*** ? the one forge that does native **\*\*video + audio + timestamp-grounding + GRPO/RFT with a custom video reward\*\*** in one stack. Keep **\*\*unsloth\*\*** (Apache-2.0 core; *\*avoid its AGPL Studio UI\**) as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-time snapshot ? re-verify at impl.
- **\*\*Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)\*\*** ? long-video context, purpose-built **\*\*timestamp emission\*\*** (second-level localization), multi-turn video chat. Re-confirm the exact size's LICENSE pre-release. **\*\*Gemma-4 stays AMBER and bench-only\*\*** ? its Apache grant is likely but its Prohibited-Use posture is unconfirmed (counsel needed), *\*and\** its video/timestamp capability is disputed; not the primary extractor.  
**\*\*InternVL3 (MIT/Apache)\*\*** = clean fallback if a Qwen blocker appears. ? base swap is cheap *\*only within timestamp-capable Apache bases\** (timestamp data-format + grounding-CoT are base-specific).
- **\*\*START WITH THE TAL HEAD, NOT THE VLM\*\*** (the decisive call): fine-tuned VLMs **\*\*miss strict temporal boundaries\*\*** (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").  
**\*\*Boundary accuracy IS the product.\*\*** The head (``rally_seq_proto.py``) is already de-risked on our corpus (held-out AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super (\$0), is ~1M params (trivially compressible later), and **\*\*doubles as a pseudo-label teacher + the honest baseline the VLM must beat.\*\*** So: **\*\*head now ? VLM later\*\***, strictly sequenced. When the VLM comes, prefer **\*\*two-stage\*\*** (VLM coarse-proposes, head refines boundaries) over wholesale replacement. *\*(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)\**

## Conversational video-QA ? build the bookkeeping NOW, punt the model

**\*\*Punt the conversational \*feature\*\*\*** until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI over a weak extractor exposes the weakness). **\*\*Reject end-to-end-VLM-answers-each-turn\*\*** (expensive; bad at exact counting; the strong teachers are license-forbidden). **\*\*But build the \*seam\* now\*\*** (the brief requires it):

- **\*\*Architecture = "extract once, query many":\*\*** EXTRACTION (TAL head ? later VLM) emits structured timestamped spans ? a durable **\*\*per-video EVENT STORE\*\*** ? a **\*\*structured QUERY layer\*\*** (text-to-SQL + ffmpeg clip-cut). The cited queries ("longest rally", "all service faults", "rallies > 20 shots") are **\*\*structured-numeric ? SQL, not a**

## 27. user (2026-06-20T10:03:47.694Z)

```
1      @import "tailwindcss";
2
3      /* Brand tokens (mirror the marketing site palette). Tailwind v4 generates utilities
4         from these ? e.g. bg-paper, text-ink, bg-ink, text-lime, text-green. */
5      @theme {
6          --color-ink: #0c261f;
7          --color-ink2: #123a2e;
8          --color-green: #19b36b;
9          --color-lime: #c7f2b0;
10         --color-paper: #f6f8f4;
11     }
12
```

## 28. user (2026-06-20T10:03:47.696Z)

```
1      <!doctype html>
2      <html lang="en">
3      <head>
4          <meta charset="utf-8" />
5          <meta name="viewport" content="width=device-width, initial-scale=1" />
6          <title>KhelSutra ? 1-page capture checklist</title>
7          <meta name="description" content="How to record a badminton session so KhelSutra gives
8 you great highlight reels: camera placement, settings, stability, lighting." />
9          <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
10 viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'/%3E%3Cpath d='M13
11 L23 16 L13 22 Z' fill='%2319B36B'/%3E%3C/svg%3E" />
12          <link rel="preconnect" href="https://fonts.googleapis.com" />
13          <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
14          <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;
15 700;800&display=swap" rel="stylesheet" />
16          <style>
17              :root{--ink:#0C261F;--green:#19B36B;--green-d:#137a4c;--paper:#F6F8F4;--tx:#14271F;--
18 mut:#566B61;--line:#E1E7DF;--font:'Plus Jakarta Sans',system-ui,-apple-system,Segoe
19 UI,Roboto,sans-serif}
20              *{box-sizing:border-box}
21              body{margin:0;font-family:var(--font);color:var(--tx);background:#E9EDE6;line-
22 height:1.5;-webkit-font-smoothing:antialiased}
23              .sheet{max-width:820px;margin:18px auto;background:#fff;border-
24 radius:16px;overflow:hidden;box-shadow:0 18px 50px -24px rgba(0,0,0,.35)}
25              .head{background:var(--ink);color:#EAF3EE;padding:22px 30px;display:flex;align-
26 items:center;justify-content:space-between;gap:1rem;flex-wrap:wrap}
27              .brand{display:flex;align-items:center;gap:.55rem;font-weight:800;font-size:1.2rem}
28              .brand .dot{color:var(--green)}
29              .brand .play{display:inline-block;width:0;height:0;border-style:solid;border-width:6px
30 0 6px 11px;border-color:transparent transparent transparent var(--green);margin:0
31 .3em;transform:translateY(1px)}
32              .head h1{font-size:1.05rem;font-weight:600;margin:0;color:#C7F2B0}
33              .body{padding:24px 30px 8px}
34              .intro{color:var(--mut);margin:0 0 16px;font-size:.98rem}
35              .rule{display:flex;gap:.7rem;background:#EAF6EF;border:1px solid #BfE6D2;border-
```

```

radius:12px;padding:14px 16px;margin-bottom:20px}
25     .rule .star{color:var(--green-d);font-size:1.3rem;line-height:1}
26     .rule b{color:var(--tx)}
27     .rule p{margin:.25rem 0 0;color:var(--mut);font-size:.92rem}
28     .grid{display:grid;grid-template-columns:1fr 1fr;gap:16px 26px}
29     .step h3{display:flex;align-items:center;gap:.5rem;font-size:1rem;font-
weight:700;margin:0 0 .4rem}
30     .step h3 .n{width:24px;height:24px;border-
radius:7px;background:var(--green);color:#04140D;display:flex;align-items:center;justify-
content:center;font-size:.85rem;flex:none}
31     .step ul{margin:0;padding-left:1.1rem}
32     .step li{margin:.18rem 0;font-size:.92rem;color:#34453d}
33     .step li b{color:var(--tx)}
34     .cols{display:grid;grid-template-columns:1fr 1fr;gap:0;border:1px solid
var(--line);border-radius:12px;overflow:hidden;margin:22px 0 8px}
35     .cols div{padding:14px 16px}
36     .cols .do{background:#EAF6EF}
37     .cols .dont{background:#Fbeeb;border-left:1px solid var(--line)}
38     .cols h4{margin:0 0 .5rem;font-size:.92rem}
39     .cols .do h4{color:var(--green-d)}
40     .cols .dont h4{color:#9a3b22}
41     .cols ul{margin:0;padding-left:1.1rem}
42     .cols li{font-size:.88rem;margin:.16rem 0;color:#3a4b43}
43     .consent{font-size:.86rem;color:var(--mut);border-top:1px solid var(--line);margin-
top:6px;padding:14px 0 4px}
44     .foot{display:flex;justify-content:space-between;flex-wrap:wrap;gap:.5rem;padding:14px
30px 24px;color:var(--mut);font-size:.85rem}
45     .foot a{color:var(--green-d);text-decoration:none}
46     @media (max-width:680px){.grid,.cols{grid-template-columns:1fr}.cols .dont{border-
left:0;border-top:1px solid var(--line)}}
47     @media print{
48         body{background:#fff}
49         .sheet{box-shadow:none;margin:0;max-width:none;border-radius:0}
50         @page{margin:12mm}
51         .noprint{display:none}
52     }
53 </style>
54 </head>
55 <body>
56 <div class="sheet">
57     <div class="head">
58         <span class="brand">Khel<span class="play" aria-hidden="true"></span>Sutra</span>
59         <h1>1-page capture checklist</h1>
60     </div>

```

## 29. assistant (2026-06-20T10:03:51.905Z)

I now have the complete picture. Let me verify the routing setup in the SPA (App.tsx has no router yet ? single screen) and check the web package.json for what's installed, plus the site server.js routing so I can give precise route recommendations. Let me also grab the precise key honesty facts from the worker-train code to be accurate about "train your own model" today.

## 30. user (2026-06-20T10:03:52.425Z)

```

1      {
2        "name": "khelsutra-web",
3        "private": true,
4        "version": "0.0.0",
5        "type": "module",
6        "description": "KhelSutra Studio ? operator console SPA (drives the rally indexer
engine).",
7        "scripts": {
8          "dev": "vite",
9          "build": "vite build",
10         "preview": "vite preview",

```



```

11     "typecheck": "tsc --noEmit",
12     "test": "vitest run"
13 },
14     "dependencies": {
15         "react": "^19.0.0",
16         "react-dom": "^19.0.0"
17     },
18     "devDependencies": {
19         "@tailwindcss/vite": "^4.0.0",
20         "@types/react": "^19.0.0",
21         "@types/react-dom": "^19.0.0",
22         "@vitejs/plugin-react": "^4.3.4",
23         "tailwindcss": "^4.0.0",
24         "typescript": "^5.7.0",
25         "vite": "^6.0.0",
26         "vitest": "^3.0.0"
27     }
28 }
29

```

### 31. user (2026-06-20T10:03:55.478Z)

```

=== site/server.js routing (static + api) ===
2:// Serves public/ and POST /api/request-access into SQLite (node:sqlite), using the
18:const PUBLIC = join(ROOT, "public");
27: ".html": "text/html; charset=utf-8", ".png": "image/png", ".svg": "image/svg+xml",
48:async function serveStatic(pathname, res) {
49:   const rel = pathname === "/" ? "index.html"
50:     : pathname === "/capture" ? "capture.html"
71:   return await serveStatic(pathname, res);
=== site/src/index.js (Worker routes) ===
2:// Request-Access form: POST /api/request-access -> validate -> insert into D1 (binding DB).
78: async fetch(request, env) {
80:   if (url.pathname === "/api/request-access") {
84:   return env.ASSETS.fetch(request); // everything else: static assets

```

### 32. user (2026-06-20T10:03:57.883Z)

```

=== worker train (real today) ===
backend/worker/__main__.py:5:   python -m backend.worker train ...           # Gen-0 train-from-
bucket (follow-up)
backend/worker/__main__.py:9:are byte-for-byte unchanged. Combine with
`RALLY_DETECTOR_IMPL=stub` and
backend/worker/__main__.py:170:   The detector trajectory is frame-indexed; the harness maps
frame->time via fps, so a WRONG
backend/worker/__main__.py:196:   """Build the trajectory DetectorRunner for `--trajectory-
source detector`, honouring
backend/worker/__main__.py:198:   stub=CPU mock). Returns the runner, or prints an error +
returns None.
backend/worker/__main__.py:201:   detector identically (one source of truth). A non-trajectory
segmenter (e.g. yolo) has
backend/worker/__main__.py:204:   from backend.pipeline.segmenters.trajectory_hybrid import
TrajectoryHybridSegmenter
backend/worker/__main__.py:214:   f"use a trajectory hybrid (wasb_hybrid /
tracknet_hybrid / fusion_hybrid).")
backend/worker/__main__.py:218:   except NotImplementedError as e:
backend/worker/__main__.py:232:   """Gen-0 train-from-bucket: pull labels (+ videos) ->
trajectories -> manifest -> shell out
backend/worker/__main__.py:233:   to training.gen0.harness (backend must NOT import training)
-> push the artifact bundle.
backend/worker/__main__.py:235:   Two trajectory sources (the train pipeline + harness are
identical either way):
backend/worker/__main__.py:236:   * ``synth`` (default) ? the CPU "mock GPU": deterministic,
label-consistent synthetic
backend/worker/__main__.py:239:   DetectorRunner (native WASB on a GPU box, or stub for a

```

```

GPU-free wiring test). This
backend/worker/__main__.py:240:         is the M3.5 "first real training" path; only the
trajectory source flips.
backend/worker/__main__.py:266:         if args.trajectory_source == "detector":
backend/worker/__main__.py:277:         print(f"[worker] trajectory source:
{args.trajectory_source}")
backend/worker/__main__.py:283:         if args.trajectory_source == "detector":
backend/worker/__main__.py:291:             # Real fps/width drive the trajectory frame->time
mapping; PROBE (don't guess) ? a
backend/worker/__main__.py:300:             print(f"[worker] detector produced no trajectory
for {name!r} ? skipping.")
backend/worker/__main__.py:302:             specs.append({"name": name, "trajectory": traj,
"labels": local_label,
backend/worker/__main__.py:305:             from backend.pipeline.detectors.stub_runner import
synth_trajectory_from_labels
backend/worker/__main__.py:307:             synth_trajectory_from_labels(local_label, traj,
fps=args.fps, frame_width=args.frame_width)
backend/worker/__main__.py:308:             specs.append({"name": name, "trajectory": traj,
"labels": local_label,
backend/worker/__main__.py:317:         src_label = "real detector" if args.trajectory_source ==
"detector" else "CPU-mock stub"
=== files ===
__init__.py
__main__.py
__pycache__
backend/api/job_worker.py
backend/worker/__init__.py
backend/worker/__main__.py

```

### 33. assistant (2026-06-20T10:04:04.008Z)

I have everything I need. The "train your own model" capability today is a CPU-mock/stub Gen-0 harness that trains a small TAL head (~1M params) on rally labels ? it is a developer/CLI path, not a user-facing Studio feature, and the VLM is roadmap. The rally-annotator (public MIT VLC plugin) is the real, shippable "annotate your footage" tool today, producing golden-label CSVs the engine ingests.

Let me do a final check: confirm the SPA has no router today (single App.tsx, no react-router dep ? confirmed in package.json) and confirm the site OG/make\_og precedent, then return the structured output.

### 34. user (2026-06-20T10:04:07.646Z)

```

=== router in web? ===
(none above = single-screen scaffold)
=== site CI deploy root ===
8:  site:
12:      working-directory: site
18:      # No deps to install (the site is dependency-free). This runs BOTH paths:
23:  web:
24:      # Operator-console SPA (web/): typecheck + unit tests + production build gate the merge.
28:      working-directory: web
35:      cache-dependency-path: web/package-lock.json
=== capture.html foot links (precedent for a sub-page) ===
8:<link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0
0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'%3E%3Cpath d='M13 10 L23 16
L13 22 Z' fill='%2319B36B'%3E%3C/svg%3E" />
9:<link rel="preconnect" href="https://fonts.googleapis.com" />
10:<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
11:<link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;8
00&display=swap" rel="stylesheet" />
127:    <span>Questions? <a href="mailto:hello@khelsutra.guru">hello@khelsutra.guru</a> . <a
href="https://khelsutra.guru">khelsutra.guru</a></span>

```

### 35. assistant (2026-06-20T10:04:11.140Z)

I have all the information needed. Let me compile the structured output mapping the khelsutra UI surface and recommendations.

### 36. user (2026-06-20T10:05:31.564Z)

Structured output provided successfully

### 37. assistant (2026-06-20T10:05:45.189Z)

Structured output delivered. Summary of what I mapped and recommended:

**\*\*Surfaces found (3, honestly disambiguated):\*\***

1. Marketing site ``site/public/`` ? plain HTML, no build; ``index.html`` + standalone ``capture.html`` (the sub-page pattern to copy). Routing is one-liner aliases in ``site/server.js:48-50``; the Worker serves any public/ file via ``env.ASSETS.fetch`` (``site/src/index.js:84``).
2. Studio SPA ``web/`` ? Vite/React 19/TS/Tailwind, but a single-screen scaffold today with NO router (react-router not installed, ``web/package.json:14-17``).
3. The cited ``docs/LOCAL_APPLIANCE_ARCHITECTURE.md`` does NOT exist in khelsutra ? the only console design doc is ``docs/PER_RALLY_SELECTION.md``; hosting docs still live in the engine repo per ``docs/README.md:10-13``.

**\*\*The honesty correction that shaped the whole recommendation:\*\*** the "annotate" half is real and shippable (rally-annotator is a live MIT public VLC plugin producing label CSVs the engine ingests), but "train your OWN model" is only partly real ? today it's a CPU-mock Gen-0 CLI head-trainer (``backend/worker/__main__.py:232``, default ``synth`` trajectory), and the conversational VLM is roadmap (``OWNED_MODEL_IMPLEMENTATION_PLAN.md``, status PLAN, "moat = the consented dataset, not the weights"). So the USP must be positioned as "your consented labels sharpen your detector," matching the site's existing "Read this honestly" boxes (``index.html:351-354``).

**\*\*Placement:\*\*** ship a new marketing page now (``site/public/own-model.html``, ``capture.html``-styled, links the public repo, VLC setup + CSV/ending-reason help, claim-tagged); design+owner-gate the SPA correction/label loop (the already-specified gated P10 "PATCH /api/segments ? A4 Rally Review" in ``PER_RALLY_SELECTION.md:122``) ? and do NOT add a "train now" button since no such endpoint exists. Track it via a new doc following ``PROJECT_DOC_TEMPLATE.md``, peer of ``PER_RALLY_SELECTION.md``.